On Formalizing Knowledge Soundness of Non-Interactive ZK-LaBRADOR

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Abstract. The LNP protocol (CRYPTO'22) is a widely adopted lattice-based zero-knowledge framework. However, its linear proof size makes it impractical for applications like revocable anonymous credential systems that require proving multiple statements over different witnesses simultaneously. To achieve succinctness, recent works propose combining LNP with LaBRADOR (CRYPTO'23)—a lattice-based succinct proof system—yielding a composed construction, which we refer to as ZK-LaBRADOR. It inherits the zero-knowledge property of LNP and the succinctness of LaBRADOR. However, its knowledge soundness has not been formally analyzed. In interactive settings, knowledge extraction can be achieved through rewinding techniques; however, many practical applications, such as anonymous credentials, require non-interactive proofs obtained via the Fiat–Shamir (FS) transform, where these techniques are ineffective, and naive composition can lead to an exponential blowup in the extractor. To address this, we provide a formal soundness analysis of ZK-LaBRADOR by leveraging the Predicate Special Soundness (PSS) framework proposed by Aardal et al. (CRYPTO'24).

Keywords: Lattices · Zero-Knowledge · Predicate Special Soundness.

1 Introduction

Zero-knowledge proofs (ZKPs) are fundamental primitives in cryptography that enable a prover to convince a verifier of the validity of a statement without revealing any additional information about the witness. They form essential building blocks for a wide range of privacy-preserving applications, including blind signatures and anonymous credentials. In the post-quantum setting, lattice-based ZKPs are among the most promising candidates for achieving both security and efficiency. The LNP protocol [12] by Lyubashevsky et al. is a widely adopted lattice-based zero-knowledge framework known for its efficiency and modular design. LNP achieves succinctness in all but the prover's final message, where the size grows linearly with the witness length. In contrast, LaBRADOR [6] provides succinct lattice-based proofs but does not inherently satisfy the zero-knowledge property.

To obtain succinct zero-knowledge proofs, Beullens and Seiler in LaBRADOR [6] propose replacing the prover's final message in LNP with a succinct proof of knowledge constructed using LaBRADOR that verifies the same checks. This composition—which we refer to as ZK-LaBRADOR—combines the zero-knowledge property of LNP with the succinctness of LaBRADOR. Such a composition is particularly useful in applications that require proving multiple statements over different witnesses. For example, in revocable anonymous credential systems [9], a user must prove possession of a valid credential over specific attributes while showing that her revocation handle is not contained in an accumulator that accumulates all the blacklisted users. In such settings, LNP proofs alone often result in impractically large proof sizes and therefore ZK-LaBRADOR provides a succinct zero-knowledge alternative.

Each individual component of ZK-LaBRADOR has well-analyzed knowledge soundness guarantee, and in the interactive setting, standard rewinding-based techniques can be used to establish the soundness of their sequential composition in a straightforward manner. However, practical instantiations generally require non-interactive proofs, obtained by the Fiat–Shamir (FS) transform [8] of the interactive protocol, and establishing knowledge soundness in this non-interactive setting is considerably more subtle and requires a more detailed analysis.

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1.1 Related Work

The Fiat-Shamir transformation converts a public-coin interactive protocol into a non-interactive one by replacing the verifier's challenges with hashes of the protocol messages. We focus on a special class of public-coin interactive protocols, i.e. 3-message Σ-protocols—that admit the *special-soundness* property. Intuitively, a 3-message public-coin interactive protocol is *k-special-sound* if there exists a polynomial-time algorithm that, on input a statement x and k accepting transcripts having the same first message but pairwise distinct challenges, outputs a valid witness w. When $k = 2$, the protocol is said to have *special-soundness*. For such protocols, *special-soundness* implies *knowledge soundness*. Moreover, *knowledge soundness* is preserved under the Fiat–Shamir transform with only a linear security loss in the random oracle model (ROM).

However, extending this analysis for multi-round protocols presents significant technical challenges. A naive approach typically leads to an exponential increase in the extractor's running time—by a factor of Q^μ, where Q denotes the number of random oracle queries made by the prover and μ is the number of rounds in the interactive protocol. To address this AFK [4] generalised the notion of *special-soundness* for multi-round protocols. Specifically, they showed that any $(2\mu + 1)$-message public-coin interactive protocol satisfying $\boldsymbol{K}$*-special-soundness*, with $\boldsymbol{K} = (k_1, k_2, \ldots, k_\mu)$, can be transformed into a non-interactive protocol in the ROM with only a linear security loss of at most $(Q + 1)$.

However, these results do not directly apply to protocols such as LNP and LaBRADOR, whose interactive variants lack K-special soundness, because this notion alone does not guarantee that the extracted opening actually corresponds to a valid witness—specifically, one that is short and satisfies the given constraints of the protocol. LNP [12] therefore formalizes the soundness of its non-interactive variant through probabilistic arguments. It extends the abstract sampling framework of AFK [4] by incorporating probabilistic checks following witness extraction. The analysis focuses on the case where each round involves two challenge values and assumes invertible challenge differences when sampling without replacement in the final round. In contrast, LaBRADOR [6] proves soundness only for the interactive version and assumes that security is heuristically preserved in the non-interactive setting.

To rigorously capture the knowledge soundness of such commit-and-open style protocols like LaBRADOR under Fiat-Shamir transformation, Aardal et al. [1] introduced a generalized framework of Predicate Special Soundness (PSS). A PSS protocol consists of two main components: (*i*) a *special-soundness structure*, which specifies how to extract valid openings from commitments, and (*ii*) a *set of predicates*, which impose additional constraints on the extracted openings and challenge values. Intuitively, when a collection of transcripts satisfies both the soundness structure and the predicates, one can extract a valid witness. To analyze the knowledge soundness of such PSS protocols, Aardal et al. extended the proof techniques of AFK [4] to account for these predicate constraints. The key insight of their analysis lies in bounding the *failure density* of the predicates, i.e. the maximal fraction of challenges for which a malicious prover without a valid witness can still satisfy the predicate conditions. They further show that, for the adaptive Fiat–Shamir transform of any PSS protocol, an efficient knowledge extractor exists whose success probability depends only on the transcript tree structure and the failure densities of the predicates.

1.2 Our Contribution

Although prior works [2,9] have employed ZK-LaBRADOR in their constructions, they typically assume the knowledge soundness of its components—LNP and LaBRADOR—without providing a formal proof for the composed construction. In this work, we address this gap by formally analyzing the soundness of ZK-LaBRADOR under the Predicate Special Soundness (PSS) framework.

Aardal et al. [1] proved that the non-interactive LaBRADOR satisfies soundness under the PSS framework. Building on their result, we formalize the knowledge soundness of LNP within the same framework. Analyzing both components under a common setting enables a modular composition of their soundness guarantees, yielding a unified and rigorous justification of the overall knowledge soundness of the non-interactive ZK-LaBRADOR.

For simplicity, we illustrate the soundness analysis of ZK-LaBRADOR for the revocable anonymous credential construction from [9] as a concrete example. We adopt the LNP layer description and the conversion of its verification equations into an instance of the LaBRADOR layer as presented in their work, and additionally provide a detailed description of the LaBRADOR layer, which was not

explicitly detailed therein. Nevertheless, the presented analysis generalizes to all relations supported by ZK-LaBRADOR.

2 Preliminaries

2.1 Notations

We use λ to denote the security parameter for all cryptographic algorithms. For any $n \in \mathbb{N}$, we define the index set as $[n] := \{1, \ldots, n\}$. Let $\mathbb{Z}_p$ denote the ring of integers modulo p. We use the notation $\vec{v}$ to represent vectors over $\mathbb{Z}_p$, and uppercase letters such as A to represent matrices over $\mathbb{Z}_p$.

For $\vec{v} \in \mathbb{Z}_q^n$ and $p \in \{1, 2, \infty\}$, we define $\|\vec{v}\|_p$ as the ℓ_p-norm of its unique representative in $[-\frac{q-1}{2}, \frac{q-1}{2}]^n$. We write $r \leftarrow D$ to denote that r is sampled uniformly at random from a finite set or according to distribution D.

For q an odd prime and d a power of two, let $\mathcal{R}$ and $\mathcal{R}_q$ denote the polynomial rings $\mathbb{Z}[X]/\langle X^d+1\rangle$ and $\mathbb{Z}_q[X]/\langle X^d+1\rangle$, respectively. We use bold letters $\mathbf{r} \in \mathcal{R}$, $\mathbf{s} \in \mathcal{R}_q$ to denote polynomials in these rings. This bold notation extends to vectors and matrices as well. For example, $\vec{\mathbf{v}} \in \mathcal{R}_q^n$ and $\mathbf{A} \in \mathcal{R}_q^{m\times n}$ denote a vector and a matrix over $\mathcal{R}_q$, respectively. We consider all vectors as column vectors by default.

For a polynomial $\mathbf{r} = \sum_{i=0}^{d-1} r_i X^i \in \mathcal{R}_q$, we use $\mathrm{ct}(\mathbf{r})$ to denote its constant coefficient r_0. Let $\tau : \mathcal{R}_q \to \mathbb{Z}_q^d$ denote the coefficient embedding that maps each ring element to its d-dimensional coefficient vector over $\mathbb{Z}_q$. Under this embedding, the notion of norms over $\mathbb{Z}_q$ naturally extends to the polynomial ring $\mathcal{R}_q$. Unless stated otherwise, we use $\|\cdot\|$ to denote the ℓ_2-norm. For a non-negative integer α, we define the bounded-norm set $S_\alpha = \{\mathbf{x} \in \mathcal{R} \mid \|\mathbf{x}\|_\infty \le \alpha\}$. We also define $\mathcal{B}_1$ as the set of all polynomials in $\mathcal{R}_q$ whose coefficients lie in $0, 1$. Finally, for any element $\mathbf{r} \in \mathcal{R}_q$, we define its operator norm as $\|\mathbf{r}\|_{\mathrm{op}} = \max_{\mathbf{s}\in\mathcal{R}_q} \frac{\|\mathbf{rs}\|}{\|\mathbf{s}\|}$.

2.2 Cyclotomic Rings

Let d be a power of two, and $l \in \mathbb{Z}$ such that $l \mid d$ and $q \equiv 2l + 1 \pmod{4l}$. This condition ensures that $\mathbb{Z}_q$ contains a primitive $2l$-th root of unity ζ, but no element of order a higher power of 2. Then, in modulo q, the polynomial $X^d + 1$ factors as the product of l irreducible polynomials $X^\delta - \zeta_i$, each of degree $\delta = d/l$, where $\zeta_i = \zeta^{2i-1}$ for $i \in [l]$ [13, Cor. 1.2]. We refer to l as the *split factor* and δ as the *split ratio* of the ring $\mathcal{R}_q$. By the Chinese Remainder Theorem (CRT), it follows that the ring $\mathcal{R}_q = \mathbb{Z}_q[X]/\langle X^d + 1\rangle$ splits into a product of l residue fields.

2.3 Automorphisms

Let $\mathrm{Aut}(\mathcal{R}_q)$ denote the automorphism group of $\mathcal{R}_q$, where each $\sigma_i \in \mathrm{Aut}(\mathcal{R}_q)$ is defined by $\sigma_i(X) = X^i$. In this paper, we use the notation σ to refer specifically to the conjugation automorphism σ_{-1}, defined as $\sigma_{-1}(X) = X^{-1}$. This automorphism is of special interest as it relates the inner product of two vectors in $\mathcal{R}_q^n$ to the inner product of their coefficient embeddings over $\mathbb{Z}_q^{nd}$. Specifically, for $\vec{\mathbf{x}}, \vec{\mathbf{y}} \in \mathcal{R}_q^n$, let $\tau(\vec{\mathbf{x}}), \tau(\vec{\mathbf{y}}) \in \mathbb{Z}_q^{nd}$ denote their coefficient embeddings, then $\langle \tau(\vec{\mathbf{x}}), \tau(\vec{\mathbf{y}})\rangle = \mathrm{ct}\left\langle \sigma(\vec{\mathbf{x}}), \vec{\mathbf{y}}\right\rangle$.

2.4 Lattices

A lattice $\mathcal{L}$ is a discrete additive subgroup of $\mathbb{R}^n$ defined as the set of all integer linear combinations of a given basis of linearly independent vectors. Formally, for a basis $B = \{b_1, b_2, \ldots, b_m\} \subset \mathbb{R}^n$, the corresponding lattice is

$$\mathcal{L}(B) = \left\{ \sum_{i=1}^{m} z_i b_i \mid z_i \in \mathbb{Z} \right\}.$$

Here, m is the rank of the lattice, and if $m = n$, the lattice is full-rank. For any vector $\vec{y} \in \mathbb{R}^n$, the *coset* of $\mathcal{L}$ corresponding to $\vec{y}$ is defined as

$$\vec{y} + \mathcal{L} = \{\vec{y} + \vec{x} \mid \vec{x} \in \mathcal{L}\}.$$

Definition 1 (Module-SIS$_{m,n,\mathcal{B}}$ [10,12]). *Let $m, n > 0$ and $\mathcal{B} < q$ be a positive number. The Module-SIS problem over $\mathcal{R}_q$ is defined as – given a matrix $\mathbf{P} \in \mathcal{R}_q^{m \times n}$, find a nonzero vector $\vec{\mathbf{s}} \in \mathcal{R}_q^n$ such that $\mathbf{P}\vec{\mathbf{s}} = \mathbf{0}$ and $0 < \|\vec{\mathbf{s}}\| \leq \mathcal{B}$. The adversary $\mathcal{A}$'s advantage in solving MSIS is ε if*

$$\Pr\left[0 < \|\vec{\mathbf{s}}\| \leq \mathcal{B} \ \wedge \ \mathbf{P}\vec{\mathbf{s}} = \mathbf{0} \ \middle| \ \mathbf{P} \in \mathcal{R}_q^{m \times n}, \ \vec{\mathbf{s}} \leftarrow \mathcal{A}(\mathbf{P})\right] \geq \varepsilon.$$

The MSIS-Assumption states that no probabilistic polynomial time (PPT) adversary can solve this problem with non-negligible advantage.

2.5 Probability Distributions

Discrete Gaussian Distribution. In an m-dimensional space, the discrete Gaussian over a lattice coset $\vec{y} + \mathcal{L}$ with center $\vec{c}$ and standard deviation $\mathfrak{s} > 0$ is defined by

$$D_{\vec{y}+\mathcal{L},\mathfrak{s},\vec{c}}(\vec{x}) = \frac{\exp\left(-\|\vec{x} - \vec{c}\|^2/(2\mathfrak{s}^2)\right)}{\sum_{\vec{x}' \in \vec{y}+\mathcal{L}} \exp\left(-\|\vec{x}' - \vec{c}\|^2/(2\mathfrak{s}^2)\right)}, \quad \forall \vec{x} \in \vec{y} + \mathcal{L}.$$

Lemma 1 (Modular Johnson-Lindenstrauss Lemma [1]). *Let C be the distribution over $\{-1, 0, 1\}$ where 0 occurs with probability $\frac{1}{2}$ and each of ± 1 occurs with probability $\frac{1}{4}$. Let $q \in \mathbb{N}$. For every $\lambda \in \mathbb{N}$, There exist constants $C^{(1)} = C^{(1)}(\lambda)$ and $C^{(2)} = C^{(2)}(\lambda)$ such that the following holds:*

For every vector $\vec{x} \in \mathbb{Z}_q^n$ with $\|\vec{x}\| \geq \mathcal{B}$, where the bound $\mathcal{B} \leq q/C^{(1)}$,

$$\Pr_{\Pi \leftarrow \mathsf{C}^{2\lambda \times n}}\left[\|\Pi\vec{x} \bmod q\| < \sqrt{C^{(2)}}\mathcal{B}\right] \lesssim 2^{-\lambda}.$$

Here, $\lesssim$ denotes that this result is heuristic in nature.

2.6 Challenge Space

We define $\mathcal{C} \subset \mathcal{R}_q$ as the challenge space, such that for any distinct $\mathbf{c}_1, \mathbf{c}_2 \in \mathcal{C}$, the difference $\mathbf{c}_1 - \mathbf{c}_2$ is invertible. Moreover, each $\mathbf{c} \in \mathcal{C}$ satisfies $\sqrt[2k]{\|c^{2k}\|} \leq \eta$ and $\|\mathbf{c}\|_{\text{op}} \leq T_{\text{op}}$, for some constants $\eta, T_{\text{op}} \in \mathbb{R}$.

A subset $\mathcal{C} \subset \mathcal{R}_q$ is said to be B-well-spread if, for every index $i \in [l]$ and any element $\boldsymbol{w}$ in the quotient ring $\mathbb{Z}_q[X]/\langle X^\delta - \zeta_i\rangle$, the probability that a randomly chosen $\boldsymbol{c} \in \mathcal{C}$ satisfies $\boldsymbol{c} \bmod (X^\delta - \zeta_i) = \boldsymbol{w}$ is at most B.

Lemma 2 ([1]). *Let $\mathcal{C} \subset \mathcal{R}_q$ be B-well-spread for some $B \in [0, 1]$. Then, for any arbitrary element $\boldsymbol{y} \in \mathcal{R}$, the probability that the difference between a randomly sampled $\boldsymbol{c} \in \mathcal{C}$ and $\boldsymbol{y}$ is invertible in $\mathcal{R}_q$ is at least $1 - lB$, i.e.*

$$\Pr_{\boldsymbol{c} \leftarrow \mathcal{C}}\left[\boldsymbol{c} - \boldsymbol{y} \in \mathcal{R}_q^\times\right] \geq 1 - lB.$$

This implies that when B is negligible, the likelihood of non-invertible challenge differences is also negligible.

2.7 Approximate Range Proofs

In certain analyses, it suffices to show that coefficients of a vector are small rather than tightly bounding its norm. This is particularly useful for ensuring that the inner product of vectors does not wrap around modulo q, i.e. it holds over the integers. The intuition behind these proofs follows from the observation that the inner product (modulo q) of a random vector $\vec{r} \leftarrow \text{Bin}_m$ with an arbitrary vector $\vec{w} \in \mathbb{Z}_q^m$ is smaller than $\frac{1}{2}\|\vec{w}\|_1$ with probability at most $\frac{1}{2}$ [5]. The following lemma, adapted from LNP [12], provides a more general bound in terms of the ℓ_2-norm [12].

Lemma 3. *Fix $m, P \in \mathbb{N}$ and a bound $b \leqslant P/41m$, and let $\vec{v} \in [\pm P/2]^m$ with $\|\vec{w}\| \geqslant b$, and let $\vec{y} \in [\pm P/2]^m$. Then*

$$\text{Pr}_{\mathfrak{R} \leftarrow \text{Bin}_1^{256 \times m}}\left[\|\mathfrak{R}\vec{v} + \vec{y} \bmod P\| < \frac{1}{2}b\sqrt{26}\right] < 2^{-128}.$$

2.8 Interactive proofs

Let $\mathrm{R} \subseteq \{0,1\}^* \times \{0,1\}^*$ be a binary relation, where each pair $(\mathrm{x}, \mathrm{w}) \in \mathrm{R}$ consists of a *statement* x and a corresponding *witness* w. We focus on NP relations—those for which membership $(\mathrm{x}, \mathrm{w}) \in \mathrm{R}$ can be verified in time $\mathrm{poly}(|\mathrm{x}|)$. In particular it follows that $|\mathrm{w}| = \mathrm{poly}(|\mathrm{x}|)$. The associated language and witness set are defined as $L_{\mathrm{R}} = \{\, \mathrm{x} : \exists \mathrm{w} \text{ s.t. } (\mathrm{x}, \mathrm{w}) \in \mathrm{R} \,\}$, $\mathrm{R}(\mathrm{x}) = \{\, \mathrm{w} : (\mathrm{x}, \mathrm{w}) \in \mathrm{R} \,\}$. We refer to [3,4] for the definitions presented below.

An interactive proof for a relation R is a two-party protocol $\Pi = (\mathcal{P}, \mathcal{V})$ between a prover $\mathcal{P}$ and a verifier $\mathcal{V}$, both modeled as PPT algorithms. The prover aims to convince the verifier that it knows a witness w for a public statement x. Both parties receive x as input, while $\mathcal{P}$ additionally holds a private witness $\mathrm{w} \in \mathrm{R}(\mathrm{x})$. The execution on inputs $(\mathrm{x}; \mathrm{w})$ is denoted $\Pi(\mathrm{x}; \mathrm{w})$, after which $\mathcal{V}$ outputs either accept or reject. The sequence of exchanged messages forms the transcript, which is accepting if $\mathcal{V}$ outputs accept. An interactive proof with μ communication rounds is called a $2\mu + 1$-message protocol. If the verifier's randomness is public, it is called a public-coin $2\mu + 1$-message protocol. By default, the communication transcript is represented as $(a_1, c_1, \ldots, a_\mu, c_\mu, a_{\mu+1})$.

We define the standard security properties for an interactive protocol Π:

1. *Completeness.* Π is perfectly complete if an honest verifier accepts every valid statement-witness pair $(\mathrm{x}, \mathrm{w}) \in \mathrm{R}$ with probability 1. In a relaxed notion, an honest execution may be rejected with a negligible probability, referred to as the *completeness error*.
2. *Knowledge Soundness.* Π is knowledge sound if any prover convincing the verifier on $\mathrm{x} \in L_{\mathrm{R}}$ with non-negligible probability must "know" a valid witness $\mathrm{w} \in \mathrm{R}(\mathrm{x})$. Formally, there exists an efficient knowledge extractor that can produce such a witness from any successful prover.
3. *Proof/Argument of Knowledge.* Π is a proof of knowledge (PoK) if it satisfies both completeness and knowledge soundness. If knowledge soundness holds only against polynomial-time provers, it is called an argument of knowledge (AoK).
4. *Honest-Verifier Zero Knowledge (HVZK).* Π is HVZK if a polynomial-time simulator on input $\mathrm{x} \in L_R$ produces accepting transcripts statistically close to those of honest executions.

***K*-Special Soundness.** Let $(k_1, \ldots, k_\mu)$- *tree of transcripts* be a collection of $K = \prod_{i=1}^{\mu} k_i$ accepting transcripts arranged in a tree where each node at depth i corresponds to the prover's message and has k_i children, each representing a distinct verifier challenge. A $(2\mu+1)$-message public-coin protocol is said to be $\boldsymbol{K} = (k_1, \ldots, k_\mu)$*-special sound* if there exists an efficient algorithm that, given a $(k_1, \ldots, k_\mu)$-tree of transcripts, can extract a witness $\mathrm{w} \in \mathrm{R}(\mathrm{x})$ from this tree.

2.9 Non-Interactive Random Oracle Proofs

The Fiat-Shamir [8] transformation converts a (public-coin) interactive proof into a non-interactive random oracle proof (NIROP) by replacing the verifier's random challenges with values derived from a random oracle. For a $(2\mu+1)$-message protocol, the i-th challenge is defined as $c_i = \mathsf{RO}_i(a_1, \ldots, a_i)$, for each i we assume the range of RO_i to be the challenge set $\mathcal{C}_i$ and its domain to be $\{0,1\}^*$, i.e. $\mathsf{RO}_i : \{0,1\}^* \to \mathcal{C}_i$.

The *completeness* of the FS[Π] follows from the correctness of the underlying interactive protocol. The *HVZK* property also extends naturally since the prover omits aborting transcripts. However, for *knowledge soundness*, the error depends on the number of queries that a cheating prover is allowed to make to the RO. This leads to the corresponding notion of knowledge soundness for NIROPs, analogous to that in interactive proofs.

Definition 2 (Knowledge Soundness – NIROP). *A non-interactive random oracle proof $(\mathcal{P}, \mathcal{V})$ for a relation* R *is knowledge-sound with knowledge error κ if there exist a polynomial p and a polynomial-time algorithm $\mathcal{E}$ (the knowledge extractor) such that, for any Q-query prover $\mathcal{P}^*$ and all* $\mathrm{x} \in \{0,1\}^*$*,*

$$\Pr[\mathrm{w} \in R(\mathrm{x}) : \mathrm{w} \leftarrow \mathcal{E}^{\mathcal{P}^*}(\mathrm{x})] \geq \frac{\epsilon(\mathcal{P}^*, \mathrm{x}) - \kappa(|\mathrm{x}|, Q)}{p(|\mathrm{x}|)},$$

where $\epsilon(\mathcal{P}^*, \mathrm{x}) = \Pr[\mathcal{V}^{\mathsf{RO}}(\mathrm{x}, \mathcal{P}^*, \mathsf{RO}) = 1]$. Here, $\mathcal{E}$ has oracle access to $\mathcal{P}^*$ and may program RO *arbitrarily. The probability is over the randomness of $\mathcal{E}$, $\mathcal{V}$, $\mathcal{P}^*$ and* RO*.*

2.10 Predicate Special Soundness (PSS)

The main technical challenges in adapting the AFK framework [4] of $\boldsymbol{K}$-special soundness for FS transformed multi-round interactive protocols to lattice-based protocols like LNP and LaBRADOR are as follows.

1. *Bad challenges.* In lattice-based settings, the challenge space may be imperfect— i.e., not every challenge difference is invertible in $\mathcal{R}_q$. Consequently, extraction may fail for such *bad challenges*. To address this issue, LNP [12] considers a challenge space with always invertible challenge differences.
2. *Probabilistic extraction.* In protocols like LNP and LaBRADOR, the extractor only outputs a candidate witness, which requires additional probabilistic checks to verify its validity.
3. *Vector of challenges.* AFK only supports extraction from trees labelled by single challenges, whereas LaBRADOR requires trees labelled by vector of challenges that must be coordinate-wise distinct for extraction.

To address issues (1) and (2), the Predicate Special Soundness (PSS) framework extends $\boldsymbol{K}$-special soundness with a predicate system $\boldsymbol{\Phi}$, yielding the $(\boldsymbol{K}, \boldsymbol{\Phi})$-PSS framework. Here, $\boldsymbol{\Phi}$ defines level-wise predicates over the $\boldsymbol{K}$-tree to ensure bottom-up well-formedness of transcript subtrees. It introduces two types of predicates: a) *challenge predicates*—enforcing special properties (e.g. invertibility) on sibling challenges, and b) *commitment predicates*—ensuring consistency of committed values for witness extraction. Each predicate has a *failure density* that bounds the fraction of bad challenges in any fixed context. These bounds give a concrete knowledge error for Fiat–Shamir-transformed $(\boldsymbol{K}, \boldsymbol{\Phi})$-PSS protocols, which, as in the AFK framework, remains linear in Q. To address issue (3), the framework further incorporates a *coordinate-wise* extension of PSS.

We recall a few key concepts from [1] that are used to analyze PSS protocols.

Definition 3 (Tree of Transcripts). *Let $\mu, k_1, \ldots, k_\mu \in \mathbb{N}$. Consider a $(2\mu+1)$-message public-coin proof (or argument) of knowledge Π for a relation* R*. Additionally, let level $m \in [\mu]$ and challenge index $\ell \in [k_m]$.*

- *We define $\mathbb{T}_{\mu+1}$ as the set of all possible accepting transcripts for Π.*
- *We define $\mathbb{T}^{(\ell)}_{m+1}$ as the set of accepting $(1, \ldots, 1, \ell, k_{m+1}, \ldots, k_\mu)$-trees of transcripts, where $\mathbb{T}^{(k_m)}_{m+1} = \mathbb{T}_m$. Each $t \in \mathbb{T}^{(\ell)}_{m+1}$ is a tuple $t = (t_1, \ldots, t_\ell)$.*
- *For $t \in \mathbb{T}_{m+1}$, we define $\mathsf{chal}_i(t)$ for $i \in [m]$ to be the i-th challenge $c_i \in \mathcal{C}_i$ shared by the transcripts.*
- *Let $C^{(l)}_m$, a tuple of the form $(c_1, \ldots, c_\ell)$, denote the challenge set at the m-th level, where $c_i \neq c_j$ for all $i \neq j$.*

2.10.1 Challenge Predicates. They enforce structured relationships on challenges beyond distinctness. For a level $m \in [\mu]$ and challenge index $\ell \in [k_m]$, a challenge predicate is a polynomial-time computable function $\Phi^{\text{chal}}_{m,\ell} : \mathcal{C}^{(\ell)}_m \to \{0, 1\}$. These predicates are applied sequentially to ensure that each challenge satisfies specific properties determined by preceding challenges. Collectively, these predicates define the structural constraints necessary for valid openings, enabling witness extraction. For instance, a witness can be extracted from two openings when the difference between their challenges is invertible. However, efficient extraction is not guaranteed for all predicates; it holds only for predicates with bounded failure density.

Definition 4 (Failure density of a challenge predicate). *Let level $m \in [\mu]$ and challenge index $\ell \in [k_m]$. Given $\overrightarrow{c_m} = (c_1, \ldots, c_{\ell-1}) \in \mathcal{C}^{(\ell-1)}_m$, where the challenges satisfy $\Phi^{chal}_{m,i}(c_1, \ldots, c_i) = 1$ for all $i \in [\ell - 1]$. Let us define the set of possible ℓ-th challenges where $\Phi^{chal}_{m,\ell}$ fails as*

$$\text{BadChallenge}^{\ell}_m(\overrightarrow{c_m}) = \left\{ c \in \mathcal{C}_m \setminus \{c_1, \ldots, c_{\ell-1}\} \mid \Phi^{chal}_{m,\ell}(\overrightarrow{c_m}, c) = 0 \right\}.$$

The challenge predicate $\Phi^{chal}_{m,\ell}$ is said to have a failure density of $p^{chal}_{m,\ell}$ if it always satisfies: $\left|\text{BadChallenge}^{\ell}_m(\overrightarrow{c_m})\right| \leq p^{chal}_{m,\ell} |\mathcal{C}_m|$. This ensures that the number of invalid challenges is bounded by a fraction $p^{chal}_{m,\ell}$ of the total challenge space $\mathcal{C}_m$.

2.10.2 Commitment Predicates. They enforce properties on the committed witnesses. A commitment predicate at level m for the ℓ-th challenge is a pair of polynomial-time computable functions

$$\Phi_m^{\text{prop}} : \mathbb{T}_{m+1}^{(k_m-1)} \to \{0,1\} \quad \text{and} \quad \Phi_m^{\text{bind}} : \mathbb{T}_{m+1}^{(k_m-1)} \times \mathbb{T}_{m+1} \to \{0,1\},$$

where Φ_m^{prop} and Φ_m^{bind} denote the property and binding predicate, respectively. The property predicate enforces desired constraints on the witness (e.g., shortness), while the binding predicate ensures transcript consistency, violation of which implies solving a hard problem such as M-SIS. The failure density of property predicates determines why a prover cannot succeed without satisfying the property predicate and to define its failure density, we first recall the following definitions.

Predicate System. It is a collection of predicates that defines the validity of a complete transcript tree.

The validity is defined recursively, starting from single accepting transcripts, which are considered valid, and applying the predicates bottom-up. The predicates are enforced by grouping transcripts that fork only in the last challenge into a new $(1,\ldots,1,k_\mu)$-subtree, which is valid if all level-μ predicates hold.

Definition 5 (Predicate System). *A predicate system $\boldsymbol{\Phi}$ for $\boldsymbol{K} = (k_1,\ldots,k_\mu)$ -tree assigns one commitment predicate $(\Phi_m^{prop}, \Phi_m^{bind})$ and k_m challenge predicates $\Phi_{m,1}^{chal},\ldots,\Phi_{m,k_m}^{chal}$, to each level $m \in [\mu]$. Let $\boldsymbol{\Phi}_m$ be Boolean functions defined recursively to denote whether a partial tree of transcripts satisfies the predicate system. For a single accepting transcript $t \in \mathbb{T}_{\mu+1}$: $\Phi_{\mu+1}(t) = 1$. For any subtree $t = (t_1,\ldots,t_{k_m}) \in \mathbb{T}_m$ with challenges $c_i = \mathsf{chal}_i(t)$, $\boldsymbol{\Phi}_m(t) = 1$ iff*

$$\Big(\textstyle\bigwedge_{i\in[k_m]} \boldsymbol{\Phi}_{m+1}(t_i) \wedge \Phi_{m,i}^{\text{chal}}(c_1,\ldots,c_i)\Big) \wedge \Phi_m^{prop}(t_1,\ldots,t_{k_m-1}) \wedge \Phi_m^{bind}\big((t_1,\ldots,t_{k_m-1}),t_{k_m}\big) = 1.$$

Binding Relation. To enforce properties on openings, commitment predicates rely on the binding property of commitments. If a prover could change its opening according to the challenges, it might produce accepting transcripts without knowing a valid witness. To prevent this, the security is reduced to the hardness of the binding problem – the extractor must therefore output either a valid witness for the original relation R_{pp} or a solution to the binding problem defined by binding relation R_{bind}.

Definition 6 (Binding Relation). *Let Π be a $(\boldsymbol{K},\boldsymbol{\Phi})$-predicate-special-sound argument of knowledge for a relation* R *and let* R_{bind} *be an additional relation. The predicate system $\boldsymbol{\Phi}$ is said to admit* R_{bind} *as a binding relation if there exists a polynomial-time algorithm $\mathcal{B}$ such that, for any level $m \in [\mu]$, statement x, and transcript tree $(t_1,\ldots,t_{k_m}) \in \mathbb{T}_m$ satisfying $\Phi_m^{\mathsf{bind}}((t_1,\ldots,t_{k_m-1}),t_{k_m}) = 0$ and*

$$\forall i \in [k_m] : \Phi_{m+1}(t_i) = 1 \quad \textit{and} \quad \Phi_{m,i}^{\mathsf{chal}}(\mathsf{chal}_m(t_1),\ldots,\mathsf{chal}_m(t_i)) = 1,$$

it holds that $\mathcal{B}(t_1,\ldots,t_{k_m}) \in \mathrm{R}_{bind}(x)$.

We now define the failure density of commitment predicates below.

Definition 7 (Failure density of a commitment predicate). *Let level $m \in [\mu]$, the set of bad subtrees is defined as*

$$\mathsf{PropUnsat}_m = \left\{ (t_1,\ldots,t_{k_m-1}) \in \mathbb{T}_{m+1}^{(k_m-1)} \;\middle|\; \begin{array}{l} \forall i \in [k_m-1] : \boldsymbol{\Phi}_{m+1}(t_i) = 1, \\ \Phi_{m,i}^{\mathsf{chal}}(c_1,\ldots,c_i) = 1, \\ \Phi_m^{\mathsf{prop}}(t_1,\ldots,t_{k_m-1}) = 0 \end{array} \right\},$$

where $c_i = \mathsf{chal}_m(t_i)$. For each $(t_1,\ldots,t_{k_m-1}) \in \mathbb{T}_{m+1}^{(k_m-1)}$, the set of subtrees that satisfy the binding predicate and the other constraints is defined as

$$\mathsf{BindTree}_m(t_1,\ldots,t_{k_m-1}) = \left\{ t \in \mathbb{T}_{m+1} \;\middle|\; \begin{array}{l} (t_1,\ldots,t_{k_m-1},t) \in \mathbb{T}_m,\ \Phi_{m+1}(t) = 1, \\ \Phi_{m,k_m}^{\mathsf{chal}}(c_1,\ldots,c_{k_m-1},\mathsf{chal}_m(t)) = 1, \\ \Phi_m^{\mathsf{bind}}((t_1,\ldots,t_{k_m-1}),t) = 1 \end{array} \right\}.$$

Consider the m-th level challenges associated with some subtree in the above set as $\mathsf{BindChal}_m(t_1,\ldots,t_{k_m-1}) = \{\mathsf{chal}_m(t) \mid t \in \mathsf{BindTree}_m(t_1,\ldots,t_{k_m-1})\}$. *The commitment predicate $(\Phi_m^{\mathsf{prop}},\Phi_m^{\mathsf{bind}})$ is said to have* failure density p_m^{com} *if* $\big|\mathsf{BindChal}_m(t_1,\ldots,t_{k_m-1})\big| \le p_m^{\mathsf{com}} |\mathcal{C}_m|$, $\forall (t_1,\ldots,t_{k_m-1}) \in \mathsf{PropUnsat}_m$.

2.10.3 Knowledge soundness. Now we define the notion of Predicate Special Soundness (PSS) and knowledge soundness of PSS protocols.

Definition 8 (Predicate Special Soundness). *Consider a public-coin argument of knowledge Π with $2\mu+1$ messages for a relation* R*, where the challenge space at the m-th round is $\mathcal{C}_m$. The protocol Π satisfies $(\boldsymbol{K}, \boldsymbol{\Phi})$-PSS for a parameter vector $\boldsymbol{K} = (k_1, \ldots, k_\mu)$ and a predicate system $\boldsymbol{\Phi}$ if there exists a polynomial-time algorithm such that given a $\boldsymbol{K}$-tree of transcripts t with $\boldsymbol{\Phi}(t) = 1$, it extracts a witness w where $(x, w) \in \mathrm{R}$.*

Theorem 1. *[1, Theorem.G.1]* [1] *Let Π be a $(\boldsymbol{K}, \boldsymbol{\Phi})$-predicate-special-sound argument of knowledge for a relation R_{pp} for public parameters* pp*. Furthermore, let $\mathrm{R}_{\mathrm{pp}}^{\mathrm{bind}}$ denote a binding relation for Φ with statement x. Then, the adaptive Fiat–Shamir transformation $FS[\Pi]$ is adaptively knowledge sound for the relation $\mathrm{R}_{\mathrm{pp}} \cup \mathrm{R}_{\mathrm{pp}}^{\mathrm{bind}}$, with knowledge error*

$$\varepsilon_{\mathrm{know}} = 2(Q+1) \sum_{i=1}^{\mu} \max\left(\frac{k_i - 1}{|C_i|}, p_i^{\mathrm{com}} + \sum_{\ell=1}^{k_i} p_{i,\ell}^{\mathrm{chal}} \right),$$

where Q denotes the number of random oracle queries made by the prover. The expected number of interactions the knowledge extractor requires with the prover is at most $K + Q(K-1)$, where $K = \prod_{i=1}^{\mu} k_i$.

2.10.4 Coordinate-wise PSS (CWPSS). In many lattice-based commit-and-open protocols, such as LaBRADOR [6], the prover sends only an amortized opening of multiple witness vectors. Consequently, the standard Predicate Special Soundness (PSS) framework can be extended to a more general notion, called Coordinate-Wise Predicate Special Soundness (CWPSS), as introduced in Appendix G of [1].

In CWPSS, a permissible set of challenge vectors is defined to enable efficient extraction. For each $i \in [r]$, we define a relation $\equiv_i$ between challenge vectors that differ only in their i-th coordinate. Formally, for $\vec{c}, \vec{c}' \in S^r$ and $i \in [r]$, where r denotes the number of coordinates (or witnesses),

$$\vec{c} \equiv_i \vec{c}' \iff \left(c_i \neq c_i'\right) \ \wedge \ \forall j \in [r] \setminus \{i\} : \ c_j = c_j'.$$

Let $K = r(k-1)+1$, where k denotes the number of distinct challenge vectors and r is the number of coordinates in each vector. The permissible set is defined as

$$\mathrm{SS}(S, r, k) = \left\{ \{\vec{c_1}, \ldots, \vec{c_K}\} \subseteq 2^{(S^r)} \;\middle|\; \begin{array}{l} \exists e \in [K], \forall i \in [r], \\ \exists J = \{j_1, \ldots, j_{k-1}\} \subseteq [K] \setminus \{e\}, \\ \forall j, j' \in J \cup \{e\},\ j \neq j' : \ \vec{c_j} \equiv_i \vec{c_{j'}} \end{array} \right\}.$$

The unique vector $\vec{c_e}$ that differs from all others in one coordinate is called the *central vector*. For each coordinate i, there exists exactly $k-1$ distinct challenge vectors that differ from $\vec{c_e}$ in only in the i-th coordinate. Hence, for each coordinate we have k distinct challenges, allowing multiple witnesses to be extracted from the corresponding accepting transcripts.

3 Soundness Analysis of LNP under PSS

In this section, we formally analyze the soundness of the non-interactive LNP protocol under the PSS framework, providing concrete bounds on the knowledge error. For simplicity, we focus on a specific instance, but the argument naturally extends to any set of relations compatible with the LNP framework.

[1] For the sake of simplicity we consider that the failure density term dominates $\frac{k_i - 1}{|\mathcal{C}_i|}$

3.1 LNP protocol

The LNP protocol offers a general framework for constructing zero-knowledge proofs of lattice relations. For example, suppose we want to prove knowledge of a secret polynomial vector $\vec{\mathbf{s}_1} \in \mathcal{R}_q^{m_1}$ such that $\mathbf{P}\vec{\mathbf{s}_1} = \vec{\mathbf{u}}$ over $\mathcal{R}_q$ and $\|\vec{\mathbf{s}_1}\| \leq \mathcal{B}$, for public matrices $\mathbf{P} \in \mathcal{R}_q^{n_1 \times m_1}$ and $\vec{\mathbf{u}} \in \mathcal{R}_q^{n_1}$. To express the norm constraint $\|\vec{\mathbf{s}_1}\| \leq \mathcal{B}$ for proof generation, the typical method is to use the *sum-of-four-squares theorem*, which ensures the existence of integers $a_1, a_2, a_3, a_4 \in \mathbb{Z}$ such that

$\mathcal{B}^2 - \|\vec{\mathbf{s}_1}\|^2 = a_1^2 + a_2^2 + a_3^2 + a_4^2$. Let $\tau(\mathbf{a}) := (a_1, a_2, a_3, a_4, 0, \ldots, 0) \in \mathbb{Z}_q^d$, then we can rewrite the relation we want to prove as

$$\left\| \begin{bmatrix} \vec{\mathbf{s}_1} \\ \mathbf{a} \end{bmatrix} \right\| = \mathcal{B}, \text{ and } \begin{bmatrix} \mathbf{P} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \vec{\mathbf{s}_1} \\ \mathbf{a} \end{bmatrix} = \mathbf{P}\vec{\mathbf{s}_1}.$$

For simplicity, we slightly abuse the notation and define $m_1 := m_1 + 1$, $\vec{\mathbf{s}_1} := (\vec{\mathbf{s}_1}, \mathbf{a}) \in \mathcal{R}_q^{m_1}$, and $\mathbf{P} := [\mathbf{P}\ \mathbf{0}] \in \mathcal{R}_q^{n_1 \times m_1}$.

We give an overview of the LNP protocol for proving the above relation in the interactive setting in Fig. 3.1. We label different rounds of LNP as "levels" as they are referred to later in the soundness analysis (Sec. 3.2). The protocol uses uniformly random public matrices $\mathbf{A}_1 \in \mathcal{R}_q^{n \times m_1}$, $\mathbf{A}_2 \in \mathcal{R}_q^{n \times m_2}$, $\mathbf{B}^{(e)} \in \mathcal{R}_q^{256/d \times m_2}$, $\mathbf{B}_g \in \mathcal{R}_q^{\lambda \times m_2}$, and vector $\vec{\mathbf{b}} \in \mathcal{R}_q^{m_2}$, which serve as public parameters for the ABDLOP commitment scheme. During the protocol, the prover computes auxiliary term $\mathbf{R}_2, \vec{\mathbf{r_1}}$, and $\mathbf{r_0}$ which are defined as

$$\mathbf{R}_2 = \begin{bmatrix} \mathbf{0} & \mathbf{0}\,\mathbf{0}\,\mathbf{0} \\ \sum_{i=1}^{\lambda} \boldsymbol{\psi}_i \cdot \delta_{i,257} & \mathbf{0}\,\mathbf{0}\,\mathbf{0} \\ \mathbf{0} & \mathbf{0}\,\mathbf{0}\,\mathbf{0} \\ \mathbf{0} & \mathbf{0}\,\mathbf{0}\,\mathbf{0} \end{bmatrix} \in \mathcal{R}_q^{2(m_1+256/d+\lambda) \times 2(m_1+256/d+\lambda)} \tag{1}$$

$$\vec{\mathbf{r}_1} = \begin{bmatrix} \sum_{i=1}^{\lambda} \sum_{j=1}^{256} \boldsymbol{\psi}_i\, \delta_{i,j}\, \sigma(\mathbf{r}_j) + \sum_{i=1}^{n_1} \boldsymbol{\psi}_{\lambda+i} \mathbf{p}_i \\ \mathbf{0} \\ \sum_{i=1}^{\lambda} \sum_{j=1}^{d} \boldsymbol{\psi}_i\, \delta_{i,j}\, X^{j-1} \\ \vdots \\ \sum_{i=1}^{\lambda} \sum_{j=1}^{d} \boldsymbol{\psi}_i\, \delta_{i,\,256/d-d+j}\, X^{j-1} \\ \boldsymbol{\psi}_1 \\ \vdots \\ \boldsymbol{\psi}_\lambda \\ \mathbf{0} \end{bmatrix} \in \mathcal{R}_q^{2(m_1+256/d+\lambda)} \tag{2}$$

$$\mathbf{r}_0 = -\sum_{i=1}^{\lambda} (\boldsymbol{\psi}_i\, \delta_{i,257}\, \mathcal{B} + \mathbf{h}_i) - \sum_{i=1}^{n_1} \boldsymbol{\psi}_{\lambda+i} \mathbf{u}_i \in \mathcal{R}_q. \tag{3}$$

During verification, the verifier checks the following bounds, defined as $B_e = 1.7\mathfrak{s}_3\sqrt{256}$, $B_1 = \mathfrak{s}_1\sqrt{2m_1d}$, and $B_2 = \mathfrak{s}_2\sqrt{2m_2d}$ (refer [7, Lem.5.5]).

3.2 Knowledge soundness

The LNP protocol is a public-key zero-knowledge argument consisting of $(2\mu + 1)$ messages, where $\mu = 4$. In this section, we present the full proof of its non-interactive knowledge soundness under PSS framework. Informally, knowledge soundness ensures the existence of an efficient extractor that can recover a valid witness from any prover that convinces the verifier. We begin by outlining the structure of the extractor for the adaptive Fiat–Shamir transformation of Π (Fig. 3.1).

Prover | **Verifier**

Level 0: Commit phase:
$\vec{\mathbf{s}_2} \leftarrow S_\nu^{m_2}, \quad \vec{\mathbf{y}}^{(e)} \leftarrow D_{\mathfrak{s}^{(e)}}^{256/d},$
$\vec{\mathbf{t}_A} := \mathbf{A}_1\vec{\mathbf{s}_1} + \mathbf{A}_2\vec{\mathbf{s}_2}, \quad \vec{\mathbf{t}}^{(e)} := \mathbf{B}^{(e)}\vec{\mathbf{s}_2} + \vec{\mathbf{y}}^{(e)}$

$\xrightarrow{\vec{\mathbf{t}_A},\ \vec{\mathbf{t}}^{(e)}}$

$\mathfrak{B}_0^{(e)}, \mathfrak{B}_1^{(e)} \leftarrow \{0,1\}^{256\times m_1 d}$

$\xleftarrow{\mathfrak{B}_0^{(e)}, \mathfrak{B}_1^{(e)}}$

Level 1: Approximate Range Proof (ARP):
$\vec{\mathbf{g}} \leftarrow \{\mathbf{x} \in \mathcal{R}_q : \check{\mathbf{x}} = 0\}^\lambda, \quad \vec{\mathbf{t}_g} := \mathbf{B}_g\vec{\mathbf{s}_2} + \vec{\mathbf{g}}$

$\mathfrak{B}^{(e)} := \mathfrak{B}_0^{(e)} - \mathfrak{B}_1^{(e)}, \quad \vec{z}^{(e)} := \mathfrak{B}^{(e)}\tau(\vec{\mathbf{s_1}}) + \tau(\vec{\mathbf{y}}^{(e)})$
Perform rejection sampling to ensure $\vec{z}^{(e)}$ does not leak any information about $\vec{\mathbf{s_1}}$

$\xrightarrow{\vec{z}^{(e)}, \vec{\mathbf{t}_g}}$

$\|\vec{z}^{(e)}\| \overset{?}{\leqslant} B_e$
$(\delta_{i,j})_{i\in[\lambda], j\in[257]} \leftarrow \mathbb{Z}_q^{\lambda\times 257}$

$\xleftarrow{\delta_{i,j}}$

Level 2: Masked randomized linear combinations of $\mathbb{Z}_q$ statements:
$$\mathbf{h}_i := \sum_{j=1}^{256} \delta_{i,j}\big(\sigma(\vec{\mathfrak{b}}_j^{(e)})^T\vec{\mathbf{s}_1} + \sigma(\vec{\mathbf{e}_j})^T\vec{\mathbf{y}}^{(e)} - z_j^{(e)}\big) + \delta_{i,257}\big(\sigma(\vec{\mathbf{s_1}})^T\vec{\mathbf{s}_1} - \mathcal{B}\big) + \mathbf{g}_i, \quad \text{for } i \in [\lambda],$$
where $\vec{\mathfrak{b}}_j^{(e)}, \vec{\mathbf{e}_j}, z_j^{(e)}$ are rows of $\mathfrak{B}$, identity matrix and $\overrightarrow{z^{(e)}}$ respectively.

$\xrightarrow{\mathbf{h}_1, \ldots, \mathbf{h}_\lambda}$

$\mathrm{ct}(\mathbf{h}_i) \overset{?}{=} 0$ for all $i \in [\lambda]$
$\vec{\boldsymbol{\psi}} = (\boldsymbol{\psi}_1, \ldots, \boldsymbol{\psi}_{\lambda+n_1}) \leftarrow \mathcal{R}_q^{\lambda+n_1}$

$\xleftarrow{\boldsymbol{\psi}_1, \ldots, \boldsymbol{\psi}_{\lambda+n_1}}$

Level 3: Amortising into one quadratic equation to prove $\sum_{i=1}^{\lambda}\boldsymbol{\psi}_i\Big(\mathbf{g}_i + \sum_{j=1}^{256}\delta_{i,j}\big(\sigma(\vec{\mathfrak{b}}_j^{(e)})^T\vec{\mathbf{s}_1} + \sigma(\vec{\mathbf{e}_j})^T\vec{\mathbf{y}}^{(e)} - z_j^{(e)}\big) + \delta_{i,257}\big(\sigma(\vec{\mathbf{s_1}})^T\vec{\mathbf{s}_1} - \mathcal{B}\big) - \mathbf{h}_i\Big) + \sum_{i=1}^{n_1}\boldsymbol{\psi}_{\lambda+i}\big(\vec{\mathbf{p}}_i^T\vec{\mathbf{s}_1} - \mathbf{u}_i\big) = \mathbf{0}$. Equivalently, this equation can be written as $\vec{\mathbf{s}}^T\mathbf{R}_2\vec{\mathbf{s}} + \vec{\mathbf{r}_1}^T\vec{\mathbf{s}} + \mathbf{r}_0 = \mathbf{0}$, where $\vec{\mathbf{s}} := \big[\vec{\mathbf{s_1}}\ \ \sigma(\vec{\mathbf{s_1}})\ \ \vec{\mathbf{m}}\ \ \sigma(\vec{\mathbf{m}})\big]$, $\vec{\mathbf{m}} := \big[\vec{\mathbf{y}}^{(e)}\ \ \vec{\mathbf{g}}\big]$ and $\mathbf{R}_2, \vec{\mathbf{r}_1}$, and $\mathbf{r_0}$ are defined as in Eqns. 1, 2, and 3.
$\mathbf{B} := \begin{bmatrix}\mathbf{B}^{(e)}\\ \mathbf{B}_g\end{bmatrix}, \quad \vec{\mathbf{t}_B} := \begin{bmatrix}\vec{\mathbf{t}}^{(e)}\\ \vec{\mathbf{t}_g}\end{bmatrix},$
$\vec{\mathbf{y}_1} \leftarrow D_{\mathfrak{s}_1}^{m_1}, \quad \vec{\mathbf{y}_2} \leftarrow D_{\mathfrak{s}_2}^{m_2}, \quad \vec{\mathbf{y}} := \big[\vec{\mathbf{y}_1}\ \ \sigma(\vec{\mathbf{y}_1})\ \ -\mathbf{B}\vec{\mathbf{y}_2}\ \ -\sigma(\mathbf{B}\vec{\mathbf{y}_2})\big]^\top.$
$\mathbf{q}_1 := \vec{\mathbf{s}}^T\mathbf{R}_2\vec{\mathbf{y}} + \vec{\mathbf{y}}^T\mathbf{R}_2\vec{\mathbf{s}} + \vec{\mathbf{r}_1}^T\vec{\mathbf{y}}.$
$\vec{\mathbf{t}_{q_1}} := \vec{\mathbf{b}}^\top\vec{\mathbf{s}_2} + \mathbf{q}_1, \quad \vec{\mathbf{w}} := \mathbf{A}_1\vec{\mathbf{y}_1} + \mathbf{A}_2\vec{\mathbf{y}_2}, \quad \mathbf{v} := \vec{\mathbf{y}}^T\mathbf{R}_2\vec{\mathbf{y}} + \vec{\mathbf{b}}^\top\vec{\mathbf{y}_2}.$

$\xrightarrow{\vec{\mathbf{t}_{q_1}}, \vec{\mathbf{w}}, \mathbf{v}}$

$\mathbf{c} \leftarrow \mathcal{C}$

$\xleftarrow{\mathbf{c}}$

Level 4: Commitment opening:
$\vec{\mathbf{z}_1} := \mathbf{c}\vec{\mathbf{s}_1} + \vec{\mathbf{y}_1}, \quad \vec{\mathbf{z}_2} := \mathbf{c}\vec{\mathbf{s}_2} + \vec{\mathbf{y}_2}.$
For $i = 1, 2$, perform rejection sampling to ensure $\vec{\mathbf{z_i}}$ does not leak any information about $\vec{\mathbf{s_i}}$.

$\xrightarrow{\vec{\mathbf{z}_1}, \vec{\mathbf{z}_2}}$

1. $\|\vec{\mathbf{z}_1}\| \overset{?}{\leqslant} B_1$, $\|\vec{\mathbf{z}_2}\| \overset{?}{\leqslant} B_2$
2. $\mathbf{A}_1\vec{\mathbf{z}_1} + \mathbf{A}_2\vec{\mathbf{z}_2} - \mathbf{c}\vec{\mathbf{t}_A} \overset{?}{=} \vec{\mathbf{w}}$
3. $\vec{\mathbf{z}}^T\mathbf{R}_2\vec{\mathbf{z}} + \mathbf{c}\vec{\mathbf{r}_1}^T\vec{\mathbf{z}} + \mathbf{c}^2\mathbf{r}_0 - \mathbf{f} \overset{?}{=} \mathbf{v}$
   where $\mathbf{f} := \mathbf{c}\mathbf{t}_{q_1} - \vec{\mathbf{b}}^\top\vec{\mathbf{z}_2}$. and
   $\vec{\mathbf{z}} := \big[\vec{\mathbf{z}_1}\ \ \sigma(\vec{\mathbf{z}_1})\ \ \mathbf{c}\vec{\mathbf{t}_B} - \mathbf{B}\vec{\mathbf{z}_2}\ \ \sigma(\mathbf{c}\vec{\mathbf{t}_B} - \mathbf{B}\vec{\mathbf{z}_2})\big]^\top$

Fig. 3.1. Interactive protocol Π for proving $\mathbf{P}\vec{\mathbf{s_1}} = \vec{\mathbf{u}}$ and $\|\vec{\mathbf{s_1}}\| \leq \mathcal{B}$.

Knowledge Extractor. Consider a deterministic adaptive Q-query random oracle prover P^* for $\mathsf{FS}[\Pi]$. After at most Q oracle interactions, P^* outputs $(\mathrm{x}, \pi, \mathsf{aux})$, where x is the statement, π the proof transcript, and aux auxiliary terms. P^* is extended to an algorithm $\mathcal{A}$ making $(Q+\mu)$ random-oracle queries to validate the proof. For each $i \in [\mu]$ where $\mu = 4$, $\mathcal{A}$ computes $c_i = \mathsf{RO}_i(I_i)$ with $I_i = (\mathrm{x}, a_1, \ldots, a_i)$, and outputs $(\vec{I}, t = (a_1, c_1, \ldots, a_5), v = \mathcal{V}^{\mathsf{RO}_i}(x, t), b, \mathsf{aux})$, where $\vec{I} = (I_1, \ldots, I_5)$ is the index vector and $b = 0/1$ indicates the predicate's validity.

For $m \in [\mu = 4]$, we define the extractor recursively through sub-extractors $\mathcal{E}_1, \ldots, \mathcal{E}_{\mu+1}$. The goal of $\mathcal{E}_m$ is to output a tree $t \in \mathbb{T}_m$ such that either $\boldsymbol{\Phi}(t) = 1$ or t encodes a witness for the binding relation $\mathrm{R}_{\mathrm{bind}}$. To achieve this, $\mathcal{E}_m$ interacts with $\mathcal{E}_{m+1}$ through an abstract sampling game that models oracle reprogramming and generates multiple valid transcripts sharing the same first message for witness extraction. We denote the witness extracted from a transcript t by $\mathcal{E}_m$ as $\mathcal{E}_m(t)$. At the lowest recursion level, $\mathcal{E}_5 = \mathcal{A}$ and all the sub-extractors preserve the query structure of $\mathcal{A}$. The final extractor $\mathcal{E}$ for $FS[\Pi]$ builds on $\mathcal{E}_1$ and employs lazy sampling to generate random-oracle responses. We refer the reader to [1, Appendix. H] for a detailed sub-extractor structure.

In the soundness analysis, we aim to lower-bound the success probability of $\mathcal{E}_m$. For each $m \in [\mu]$, we always assume that $b = 1$ for $\mathcal{E}_{m+1}$, and therefore $\mathcal{E}_m$ ignores the output b of $\mathcal{E}_{m+1}$.

Analyzing Levels. The analysis proceeds level by level, following the structure of the sub-extractors. Level 0 corresponds to the prover's initial commitments to the witnesses. We therefore analyze the protocol bottom-up across Levels 4, 3, 2, and 1, by defining predicates over these commitments. At each level, we analyze the failure density, assuming—without loss of generality—that all predicates from the preceding levels are satisfied. This approach enables us to progressively enforce stronger conditions on the extracted witness. Finally, by combining these level-wise guarantees we establish the overall knowledge soundness of the LNP protocol. We begin with an accepting transcript $t_5 \in \mathbb{T}_5$. By definition, if t_5 is an accepting transcript, the predicate $\boldsymbol{\Phi}_5(t_5)$ holds, i.e. $\boldsymbol{\Phi}_5(t_5) = 1$.

Level 4: This level reveals the commitment opening and is the only level with non-trivial challenge predicates. Consider the extracted transcript tree (t_1, t_2, t_3)
$\in \mathbb{T}_4$, with corresponding challenge values $\mathbf{c}_i$ for $i \in \{1, 2, 3\}$. First, we verify that all challenges from the first transcript are units: $\Phi_{4,1}^{\mathrm{chal}}(\boldsymbol{c}_1) = 1 \Leftrightarrow \boldsymbol{c}_1 \in \mathcal{R}_q^{\times}$. Next, we ensure that the difference between the first two challenges is invertible to enable the computation of the openings, i.e. $\Phi_{4,2}^{\mathrm{chal}}(\boldsymbol{c}_1, \boldsymbol{c}_2) = 1 \Leftrightarrow \boldsymbol{c}_2 - \boldsymbol{c}_1 \in \mathcal{R}_q^{\times}$. Using these predicates, we compute $\overline{\boldsymbol{c}} = \boldsymbol{c}_2 - \boldsymbol{c}_1$ and $\vec{\boldsymbol{w}}_i^* = \overline{\boldsymbol{c}}^{-1}(\vec{\boldsymbol{z}}_i^{(2)} - \vec{\boldsymbol{z}}_i^{(1)})$ for $i \in \{1, 2\}$. We then extract the terms committed in the BDLOP part as $\vec{\mathbf{m}}^* = (\vec{\mathbf{y}}^{(e)*}, \vec{\mathbf{g}}^*) = \mathbf{c}\,\vec{\mathbf{t}_B} - \mathbf{B}\,\vec{\mathbf{w}_2}^*$ and $\mathbf{q}_1^* = \mathbf{c}\,\vec{\mathbf{t}_{q_1}} - \vec{\mathbf{b}}^{\top}\vec{\mathbf{w}_2}^*$. For this level, we do not require a property predicate; we only define the binding predicate to ensure consistency of the extracted openings across transcripts, i.e. $\Phi_4^{\mathrm{bind}}(t_1, t_2, t_3) = 1 \Leftrightarrow \mathcal{E}_5(t_1) = \mathcal{E}_5(t_2) = \mathcal{E}_5(t_3)$. The extracted witness for $t_4 = (t_1, t_2, t_3)$ is defined as $\mathcal{E}_4(t_4) = (\vec{\mathbf{w}}_1^*, \vec{\mathbf{w}}_2^*, \vec{\mathbf{m}}^*, \mathbf{q}_1^*)$, which serves as input to the next level.

Lemma 4. *In the fourth level, $\Phi_{4,1}^{chal}$ has failure density $p_{4,1}^{chal}$ equal to lB and $\Phi_{4,2}^{chal}$ has failure density $p_{4,2}^{chal}$ equal to lB.*

Proof. We first consider the predicate $\Phi_{4,1}^{\mathrm{chal}}$, which enforces challenge invertibility. From Lemma 2, the probability that a randomly chosen $\boldsymbol{c} \in \mathcal{C}$ yields a non-invertible difference with a fixed $\boldsymbol{c}^* \in \mathcal{R}_q$ is at most lB, giving a failure probability $p_{4,1}^{\mathrm{chal}} = lB$. Similar argument applies to $\Phi_{4,2}^{\mathrm{chal}}$. Although excluding the fixed challenge $\boldsymbol{c}_1$ slightly refines the bound to $lB - \frac{1}{|\mathcal{C}|}$, we conservatively take $p_{4,2}^{\mathrm{chal}}$ as lB. Thus, the failure density of this predicate is $2lB$.

Level 3. This level combines all $\lambda + n_1$ constraints into an amortised quadratic constraint F using randomly chosen linear combinations.

$$
\begin{aligned}
F = &\sum_{i=1}^{\lambda} \boldsymbol{\psi}_i \cdot \Big(\mathbf{g}_i + \sum_{j=1}^{256} \delta_{i,j} \cdot \Big(\sigma\left(\vec{\mathfrak{b}}_j^{(e)}\right)^T \vec{\mathbf{s}_1} + \sigma\left(\vec{\mathbf{e}}_j\right)^T \vec{\mathbf{y}}^{(e)} - z_j^{(e)} \Big) \\
&+ \delta_{i,257} \cdot \Big(\sigma\left(\vec{\mathbf{s}_1}\right)^T \vec{\mathbf{s}_1} - \mathcal{B} \Big) - \mathbf{h}_i \Big) \quad + \sum_{i=1}^{n_1} \boldsymbol{\psi}_{\lambda+i} \cdot \Big(\vec{\mathbf{p}_i}^T \vec{\mathbf{s}_1} - \mathbf{u}_i \Big), \\
= &\sum_{i=1}^{\lambda+n_1} \boldsymbol{\psi}_i f^{(i)}(\vec{\mathbf{w}_1}^*, \vec{\mathbf{w}_2}^*, \vec{\mathbf{m}}^*, \mathbf{q}_1^*).
\end{aligned}
$$

Our goal is to verify that the amortized quadratic function yields $\mathbf{0}$ upon evaluation. Given $t \in \mathbb{T}_4$ satisfying $\boldsymbol{\Phi}(t) = 1$, we know that $\mathcal{E}_4$ will successfully extract the commitment openings.

$$\Phi_3^{\text{prop}}\ (t_1) = 1 \Leftrightarrow \begin{cases} (\overrightarrow{\mathbf{w}_i}^*, \overline{\mathbf{c}}_i)_{i\in[r]} \leftarrow \mathcal{E}_4\,(t_1)\,, \\ \forall f \in \mathcal{F} : f\,(\overrightarrow{\mathbf{w_1}}^*, \sigma(\overrightarrow{\mathbf{w_1}}^*)) = \mathbf{0}. \end{cases}$$

The binding property holds if the extracted witnesses are the same in each subtree. We make the same requirements for level $\ell = 1, 2, 3$:

$$\Phi_\ell^{\text{bind}}\ \left(t_1^{(\ell+1)}, t_2^{(\ell+1)}\right) = 1 \Leftrightarrow \begin{cases} (\overrightarrow{\mathbf{w}_{i,k}}^*, \overline{\mathbf{c}}_{i,k})_{i\in\{1,2\}} \leftarrow \mathcal{E}_{\ell+1}\left(t_k^{(\ell+1)}\right)\ \text{for } k \in \{1,2\}, \\ \forall i \in \{1,2\}, \overrightarrow{\mathbf{w}_{i,1}}^* = \overrightarrow{\mathbf{w}_{i,2}}^*. \end{cases}$$

Lemma 5. *The failure density of Φ_3^{com} in $\boldsymbol{\Phi}$, denoted by p_3^{com}, is equal to $q^{-d/l}$.*

Proof. Asssume Φ_3^{prop} doesn't hold, this implies there exists some $f^{(k_0)}$ for $k_0 \in [\lambda + n_1]$, such that $f^{(k_0)}(\overrightarrow{\mathbf{w}_1}^*, \overrightarrow{\mathbf{w}_2}^*, \overrightarrow{\mathbf{m}}^*, \mathbf{q}_1^*) \neq \mathbf{0}$. The evaluation of $f^{(k_0)}$ is fixed by the binding predicate Φ_4^{bind}. Since $f^{(k_0)}$ is nonzero, for the splitting factor l, there exists at least one CRT slot $i \in [l]$, where its evaluation is nonzero. For a uniformly random challenge $\boldsymbol{\psi}_{k_0} \in \mathcal{R}_q$, the product $\boldsymbol{\psi}_{k_0} f^{(k_0)}$ remains uniform in the i-th CRT slot. Hence, the probability that $F = \mathbf{0} \bmod (X^{d/l} - \zeta_i)$ is at most $q^{-d/l}$. Thus, the failure density of this predicate is at most $q^{-d/l}$.

Level 2. This level aggregates all the relations that we want to prove over $\mathbb{Z}_q$ into one. For all $i \in [\lambda]$,

$$\begin{aligned} \mathbf{h}_i &= \mathbf{g}_i + \sum_{j=1}^{256} \delta_{i,j}\Big(\sigma(\overrightarrow{\mathbf{b}_j}^{(e)})^T \overrightarrow{\mathbf{s}_1} + \sigma(\overrightarrow{\mathbf{e}_j})^T \overrightarrow{\mathbf{y}}^{(e)} - z_j^{(e)}\Big) + \delta_{i,257}\Big(\sigma(\overrightarrow{\mathbf{s}_1})^T \overrightarrow{\mathbf{s}_1} - \mathcal{B}\Big), \\ &= \mathbf{g}_i^* + \sum_{j=1}^{257} \delta_{i,j} f'^{(j)}(\overrightarrow{\mathbf{w}_1}^*, \overrightarrow{\mathbf{w}_2}^*, \overrightarrow{\mathbf{m}}^*, \mathbf{q}_1^*). \end{aligned}$$

Here, we need to ensure that our extracted witnesses satisfy the following constraint functions on the constant term,

$$\Phi_2^{\text{prop}}\ (t_1) = 1 \Leftrightarrow \begin{cases} (\overrightarrow{\boldsymbol{w}}_i^*, \overline{\mathbf{c}}_i)_{i\in[r]} \leftarrow \mathcal{E}_3\,(t_1)\,, \\ \forall f' \in \mathcal{F}' : \text{ct}\,(f'\,(\overrightarrow{\mathbf{w_1}}^*, \sigma(\overrightarrow{\mathbf{w_1}}^*))) = 0 \bmod q. \end{cases}$$

Lemma 6. *The failure density of Φ_2^{com} in $\boldsymbol{\Phi}$, denoted by p_2^{com} , is equal to $q^{-\lambda}$.*

Proof. This level combines all the constraints in $\mathbb{Z}_q$ into λ functions via random linear combinations. Assume, without loss of generality, that at least one non-zero $f'^{(k_0)}$ exists for $k_0 \in [257]$. Given uniformly sampled values $\delta_{i,k_0} \leftarrow \mathbb{Z}_q$, the term $\delta_{i,k_0} f'^{(k_0)}$ is uniformly random over $\mathbb{Z}_q$. Hence, the probability that it evaluates to zero is $1/q$, and across all λ repetitions, this becomes $q^{-\lambda}$. Thus, the proportion of $\delta_{i,j}$ values leading to zero evaluation is bounded by $q^{-\lambda}$. The failure density of this predicate is q^{λ}.

Level 1. At the initial level, we verify that the bound remains sufficiently short, aligning with the constraints of the approximate range proof (ARP), i.e.

$$\Phi_1^{\text{prop}}\ (t_1) = 1 \Leftrightarrow (\overrightarrow{\mathbf{w}}_i^*, \overline{\mathbf{c}}_i)_{i\in[r]} \leftarrow \mathcal{E}_2\,(t_1)\,, \|\overrightarrow{\mathbf{w_1}}^*\| \leq (2/\sqrt{26})B_e.$$

Lemma 7. *The failure density of Φ_1^{com} in $\boldsymbol{\Phi}$, given by p_1^{com} , is equal to $2^{-\lambda}$.*

Proof. Assume $\|\overrightarrow{\mathbf{w_1}}^*\| > (2/\sqrt{26})B_e$. From the previously defined predicates, it follows that the ARP equation has been correctly computed. Then by Lemma 3, for any vector $\overrightarrow{z}^{(e)'} \in \mathbb{Z}_q^{256}$ such that $\|\overrightarrow{z}^{(e)'}\| \leq B_e$, we have,

$$\Pr_{\left(R_0^{(e)'}, R_1^{(e)'}\right) \leftarrow \left(\{0,1\}^{2\lambda \times m_1 d}\right)^2} \left[\left(R_0^{(e)'} - R_1^{(e)'}\right)\tau(\overrightarrow{\mathbf{w}_1})^* + \tau(\overrightarrow{\mathbf{y}}^{(e)*}) = \overrightarrow{z}^{(e)'} \bmod q\right] \leqslant 2^{-\lambda}.$$

Thus, the failure density of this predicate is $2^{-\lambda}$.

Predicate Special Soundness of LNP. We now describe the predicate special soundness (PSS) and the binding relation required to establish the knowledge soundness of LNP. Consider the LNP protocol for the relation R, given as

$$R = \left\{ \left((\mathbf{P} \in \mathcal{R}_q^{n_1 \times m_1}, \vec{\mathbf{u}} \in \mathcal{R}_q^{n_1}, \mathcal{B} \in \mathbb{R}), (\vec{\mathbf{w}_1}^* \in \mathcal{R}_q^{m_1}) \right) \middle| \mathbf{P}\vec{\mathbf{w}_1}^* = \vec{\mathbf{u}} : \|\vec{\mathbf{w}_1}^*\| \leq \mathcal{B} \right\}.$$

Lemma 8. *The Π be the base LNP protocol Π as in (Fig. 3.1). Let $\boldsymbol{K} = (2, 2, 2, 3)$ and $\boldsymbol{\Phi}$ consists of commitment predicates $(\Phi_1^{com}, \Phi_2^{com}, \Phi_3^{com})$ and the challenge predicates $(\Phi_{4,1}^{chal}, \Phi_{4,2}^{chal})$. Then the protocol Π is $(\boldsymbol{K}, \boldsymbol{\Phi})$- predicate special sound for the relation* R.

Proof. Given a $t \in \mathbb{T}_1$ for a statement $(\mathbf{P}, \vec{\mathbf{u}}, \mathcal{B})$ such that $\boldsymbol{\Phi}(t) = 1$, we know from Φ_1^{prop}, Φ_2^{prop}, and Φ_3^{prop} that a witness can be computed from t such that $((\mathbf{P}, \vec{\mathbf{u}}, \mathcal{B}), (\vec{\mathbf{s}_1})) \in \mathrm{R}$.

Binding Relation. In the first round of the LNP protocol, the prover employs an ABDLOP commitment, the hardness of which is based on the M-SIS assumption (Theorem B.7 in [12]). In the case where binding is broken, we wish to extract solutions to the M-SIS problem, in particular we consider the relation,

$$\mathrm{R}^{\text{M-SIS}}_{\kappa,n,\beta^*,\mathbf{A}} = \left\{ \vec{\mathbf{v}} \mid \vec{\mathbf{v}} \in \mathcal{R}_q^n, \mathbf{A}\vec{\mathbf{v}} = \vec{\mathbf{0}} \text{ for some } \mathbf{A} \in \mathcal{R}_q^{\kappa \times n}, 0 < \|\vec{\mathbf{v}}\| \leq \beta^* \right\}.$$

Let $\mathbf{A} = [\mathbf{A}_1 \,|\, \mathbf{A}_2] \in \mathcal{R}_q^{n \times (m_1 + m_2)}$, denote the public matrix defining the commitment. Then the binding relation for LNP is defined as $R^{\text{M-SIS}}_{n, m_1+m_2, 4\eta(\beta_1^2+\beta_2^2), \mathbf{A}}$.

Theorem 2 (Adaption of [1, Theorem.G.1]). *Let Π denote the base* LNP *protocol 3.1 for the relation* R. *We consider a ring $\mathcal{R}_q$ of degree d with splitting factor l and a well-defined challenge space $\mathcal{C}$. The Fiat–Shamir transformation of Π preserves knowledge soundness for the relation $R \cup R^{\text{M-SIS}}_{n, m_1+m_2, 4\eta(\beta_1^2+\beta_2^2)}$, where R is the statement relation, and $R^{\text{M-SIS}}_{n, m_1+m_2, 4\eta(\beta_1^2+\beta_2^2)}$ is the binding relation. The knowledge error is bounded by*

$$\varepsilon_{\text{know}} = 2(Q+1)\left(2^{-\lambda} + q^{-\lambda} + q^{-d/l} + 2lB\right),$$

with Q adversarial queries. The expected number of extractor's interactions with the prover is at most $24 + 23Q$.

Proof. This follows from Theorem 1 and Lemmas 4–7, where $K = \sum_{i=1}^{4} k_i = 24$ from Lemma 8.

4 ZK-LaBRADOR

The ZK-LaBRADOR protocol, obtained by composing the LNP layer with the LaBRADOR framework, provides a succinct and efficient zero-knowledge proof system. This composition enables practical instantiations in privacy-preserving settings. We consider ZK-LaBRADOR instantiation for the revocable anonymous credentials of [9]. Anonymous credentials allow users to prove possession of credentials without revealing their identities or associated attributes. To enable efficient revocation, [9] introduces a positive dynamic accumulator that accumulates the set of revoked user handles. In such settings, the user's goal is to prove knowledge of a revocation attribute $\vec{\mathbf{x}} \in \mathcal{R}_q^{\ell}$, a membership witness $\vec{\mathbf{s}_x} \in \mathcal{R}_q^{m+\ell m'}$, and an additional short vector $\vec{\mathbf{r}} \in \mathcal{R}_q^{\ell}$, which, in the context of anonymous credentials, represents the user's attributes, that together satisfy

$$\begin{aligned} &\vec{\mathbf{x}} \in \{0,1\}^{\ell} \wedge \|\vec{\mathbf{s}_x}\| \leq \beta \wedge \vec{\mathbf{r}} \in \mathcal{B}_1^l, \\ &\mathbf{A}_1\vec{\mathbf{x}} + \mathbf{A}_2\vec{\mathbf{r}} = \vec{\mathbf{t}}, \\ &\left[\mathbf{A} \mid \mathbf{B} - \vec{\mathbf{x}}^{\top} \otimes \mathbf{G}\right] \cdot \vec{\mathbf{s}_x} = \vec{\mathbf{c}}, \end{aligned} \tag{4}$$

where $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}, \mathbf{B}, \vec{\mathbf{c}}, \vec{\mathbf{t}}$ are public parameters and the witnesses are highlighted in blue.

We show how ZK-LaBRADOR satisfies knowledge soundness for the relation in Eq. 4. This analysis naturally extends to any relation that can be expressed within the ZK-LaBRADOR framework.

4.1 Non-Interactive ZK-LaBRADOR.

We now describe the ZK-LaBRADOR protocol to prove the relations in Eq. 4. The proof builds on two layers – the LNP and LaBRADOR layers. We adopt the description of the LNP layer and its conversion to the LaBRADOR-compatible form from [9]. We then present the LaBRADOR layer in detail, outlining its proof structure and verification equations.

Let $\mathsf{RO}_i : \{0,1\}^* \to \mathcal{C}_i$ denote the random oracle used in round i. For simplicity, we use a single notation H to denote the random oracle throughout the non-interactive protocol. The common reference string (crs) of the protocol consists of uniformly random public matrices $(\bar{\mathbf{A}}_1, \bar{\mathbf{A}}_2, \bar{\mathbf{B}}_y, \bar{\mathbf{B}}_g, \bar{\mathbf{B}}_{\text{quad}}, \tilde{\mathbf{A}}, \tilde{\mathbf{B}}, \tilde{\mathbf{C}}, \tilde{\mathbf{D}})$ which are used for the ABDLOP commitments in the LNP layer and for Ajtai commitments in the LaBRADOR layer. The statement x is given by the tuple $(\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}, \mathbf{B}, \vec{\mathbf{t}}, \vec{\mathbf{c}})$.

4.1.1 LNP Layer.

We describe the LNP layer in Appendix A. For completeness, we summarize its input relation, commitment structure and verification equations below. The commitment openings are required for the soundness analysis and the verification equations serve as inputs to the LaBRADOR layer.

Input relation. In the LNP layer, proving the relation in Eq. 4 ultimately reduces to proving the knowledge of $\vec{\mathbf{x}} \in \mathcal{R}_q^{\ell}$, $\vec{\mathbf{r}} \in \mathcal{R}_q^{\ell}$, and $\vec{\mathbf{s}_x'} \in \mathcal{R}_q^{(m+\ell m')b}$ such that

$$\begin{aligned} &(\vec{\mathbf{x}}, \vec{\mathbf{r}}, \vec{\mathbf{s}_x'}) \in \mathcal{B}_1^{\ell+l+(m+\ell m')b}, \\ &\mathbf{A}_1 \vec{\mathbf{x}} + \mathbf{A}_2 \vec{\mathbf{r}} = \vec{\mathbf{t}}, \\ &\left[\mathbf{A} \mid \mathbf{B} - \vec{\mathbf{x}}^{\top} \otimes \mathbf{G}\right] \mathbf{G}_{\beta} \cdot \vec{\mathbf{s}_x'} = \vec{\mathbf{c}}, \\ &\mathbf{x}_i \cdot (\mathbf{x}_i - 1) = 0 \quad \text{for } i = 1, 2, \ldots, \ell. \end{aligned} \tag{5}$$

Commitment. The prover commits to the witness vector $\vec{\mathbf{s}_1} = \left[\vec{\mathbf{x}} \;\; \vec{\mathbf{r}} \;\; \vec{\mathbf{s}_x'}\right]$ using a short polynomial vector $\vec{\mathbf{s}_2}$ sampled uniformly at random from $\mathcal{B}_1^{m_2}$. Let $m_1 = \ell + l + (m + \ell m')b$ and $\bar{\mathbf{A}}_1 \in \mathcal{R}_q^{n \times m_1}$, $\bar{\mathbf{A}}_2 \in \mathcal{R}_q^{n \times m_2}$ be the public matrices from CRS. The commitment to $\vec{\mathbf{s}_1}$ is then computed as $\vec{\mathbf{t}_{\mathbf{A}}} := \bar{\mathbf{A}}_1 \vec{\mathbf{s}_1} + \bar{\mathbf{A}}_2 \vec{\mathbf{s}_2}$. Upon receiving a challenge $\mathbf{c} \in \mathcal{C}$ from the random oracle, the prover computes the commitment opening as

$$\vec{\mathbf{z}_1} = \begin{bmatrix} \vec{\mathbf{z}_x} \\ \vec{\mathbf{z}_r} \\ \vec{\mathbf{z}_{sx}} \end{bmatrix} = \vec{\mathbf{y}_1} + \mathbf{c}\vec{\mathbf{s}_1}, \quad \vec{\mathbf{z}_2} = \vec{\mathbf{y}_2} + \mathbf{c}\vec{\mathbf{s}_2}.$$

We use these openings during the extraction phase to extract the witnesses.

Verification. Let the prover output the proof $\pi_{\mathsf{LNP}} = (a_1, a_2, a_3, a_4, a_5) = ((\vec{\mathbf{t}_A}, \vec{\mathbf{t}_y}, \vec{\mathbf{t}_g}, \vec{\mathbf{w}}), (\vec{z_3}), (\mathbf{h}_1, \ldots, \mathbf{h}_\lambda), (\mathbf{t}_{\text{quad}}, \mathbf{v}), (\vec{\mathbf{z}}_1, \vec{\mathbf{z}}_2))$. The verifier checks the correctness of the proof by using the following equations:

$$\begin{aligned} &\|\vec{\mathbf{z}_1}\| \le \beta_1, \quad \|\vec{\mathbf{z}_2}\| \le \beta_2, \quad \|\vec{z_3}\| \le \beta_3, \\ &\bar{\mathbf{A}}_1 \vec{\mathbf{z}_1} + \bar{\mathbf{A}}_2 \vec{\mathbf{z}_2} = \vec{\mathbf{w}} + \mathbf{c}\vec{\mathbf{t}_A}, \\ &U(\vec{\mathbf{t}_y}, \vec{\mathbf{t}_g}, \mathbf{t}_{\text{quad}}, \mathbf{v}, \vec{\mathbf{z}}_1, \vec{\mathbf{z}}_2) = 0, \\ &\text{ct}(\mathbf{h}_i) = \sum_{j=1}^{256} \nu_{i,j} z_{3,j}, \forall i \in [\lambda], \end{aligned} \tag{6}$$

where $\beta_1 := \mathfrak{s}_1\sqrt{2m_1 d}, \beta_2 := \mathfrak{s}_2\sqrt{2m_2 d}$ and $\beta_3 := 1.7\mathfrak{s}_3\sqrt{256}$ (refer [7, Lem.5.5]), U denotes the verification function for the amortised quadratic relation formed during the protocol (refer Appendix.A, Eq.8) and $\nu'_{i,j}s$ are the third-round random oracle challenges.

4.1.2 LaBRADOR Layer.

We first recall the principal relation of LaBRADOR [6]. The principal relation is parameterized by a rank $n \ge 1$, a multiplicity $r \ge 1$, and a norm bound $\beta > 0$. Let $\mathcal{F} = (f^{(k)} \mid k \in [K])$ with

$$f(\vec{\mathbf{s}_1}, \ldots, \vec{\mathbf{s}_r}) = \sum_{i,j=1}^{r} \mathbf{a}_{i,j} \langle \vec{\mathbf{s}_i}, \vec{\mathbf{s}_j} \rangle + \sum_{i=1}^{r} \langle \vec{\boldsymbol{\varphi}_i}, \vec{\mathbf{s}_i} \rangle - \mathbf{b},$$

where $\mathbf{a}_{i,j}, \mathbf{b} \in \mathcal{R}_q$, $\vec{\boldsymbol{\varphi}}_i \in \mathcal{R}_q^n$, and $(\mathbf{a}_{i,j})$ is symmetric. For another family $\mathcal{F}' = (f'^{(l)} \mid l \in [L])$, with

$$\mathrm{ct}(f'(\vec{\mathbf{s}}_1, \ldots, \vec{\mathbf{s}}_r)) = \mathrm{ct}\Big(\sum_{i,j=1}^{r} \mathbf{a}'_{i,j} \langle \vec{\mathbf{s}}_i, \vec{\mathbf{s}}_j \rangle + \sum_{i=1}^{r} \langle \vec{\boldsymbol{\varphi}}'_i, \vec{\mathbf{s}}_i \rangle - b'_0\Big).$$

Then the principal relation is defined as

$$\mathrm{R} = \left\{ \left((\mathcal{F}, \mathcal{F}', \mathcal{B}), (\vec{\boldsymbol{s}}_1, \ldots, \vec{\boldsymbol{s}}_r)\right) \;\middle|\; \begin{array}{l} f(\vec{\boldsymbol{s}}_1, \ldots, \vec{\boldsymbol{s}}_r) = 0 \;\; \forall f \in \mathcal{F}, \\ \mathrm{ct}(f'(\vec{\boldsymbol{s}}_1, \ldots, \vec{\boldsymbol{s}}_r)) = 0 \;\; \forall f' \in \mathcal{F}', \\ \sum_{i=1}^{r} \|\vec{\boldsymbol{s}}_i\|_2^2 \le \mathcal{B}^2. \end{array} \right\}.$$

Translating LNP verification into LaBRADOR instance. Since the prover's final message $a_5 = (\vec{\mathbf{z}}_1, \vec{\mathbf{z}}_2)$, i.e. the commitment openings, is the most computationally expensive part, it is provided as the witness for the LaBRADOR relation, while the first four prover messages (a_1, a_2, a_3, a_4) are revealed in clear to the verifier. However, including the entire vector $\vec{\mathbf{z}}_1 = \left[\vec{\mathbf{z}}_x \;\; \vec{\mathbf{z}}_r \;\; \vec{\mathbf{z}}_{sx}\right]$ as part of the witness introduces a difficulty – the verification equations contain a quadratic term involving $\vec{\mathbf{z}}_x$ and $\vec{\mathbf{z}}_{sx}$, which cannot be directly represented as a principal relation of LaBRADOR. To avoid this, $\vec{\mathbf{z}}_x$ is also revealed in the clear.

A further challenge arises from quadratic equations involving automorphisms, such as terms of the form $\mathbf{z}^\top \sigma(\mathbf{z})$. The LaBRADOR framework does not natively support quadratic constraints combined with automorphisms. To handle these, the witness is extended with the automorphic images $\vec{\mathbf{z}}_{r'} := \sigma(\vec{\mathbf{z}}_r)$ and $\vec{\mathbf{z}}_{sx'} := \sigma(\vec{\mathbf{z}}_{sx})$, and the well-formedness of these need to be proven. Since σ acts as a signed permutation on polynomial coefficients, there exists a public permutation matrix $M \in \mathbb{Z}_q^{ld \times ld}$ such that $\tau(\vec{\mathbf{z}}_{r'}) = M\tau(\vec{\mathbf{z}}_r) \bmod q$. Thus, verifying $\vec{\mathbf{z}}_{r'} = \sigma(\vec{\mathbf{z}}_r)$ is equivalent to checking that, for each $i \in [ld]$, the constant term of $\sigma(\vec{\mathbf{e}}_i)^\top \vec{\mathbf{z}}_{r'} - \sigma(\vec{\mathbf{m}}_i)^\top \vec{\mathbf{z}}_r$ vanishes, where $\vec{\mathbf{m}}_i$ corresponds to the i-th row of M and $\vec{\mathbf{e}}_i$ is the unit vector. An identical check applies for $\vec{\mathbf{z}}_{sx'}$.

Finally, since the principal relation of LaBRADOR enforces a single norm bound, a relaxed bound is defined over the witnesses. Let $\mathcal{Z} = \{r, r', sx, sx', 2\}$, then $\sum_{i \in \mathcal{Z}} \|\vec{\mathbf{z}}_i\|^2 \le \beta'^2$, where $\beta' = \sqrt{2\beta_1 + \beta_2}$.

Input relation. Based on the output of the LNP layer, a new witness is set for the LaBRADOR proof system. Accordingly, the second and third equations are rewritten, and together with the relations to prove correctness of $\vec{\mathbf{z}}_{r'}$ and $\vec{\mathbf{z}}_{sx'}$, the input relation takes the following form

$$\begin{aligned}
&\sum_{i \in \mathcal{Z}} \|\vec{\mathbf{z}}_i\|^2 \le \beta'^2, \\
&\mathbf{A}' \begin{bmatrix} \vec{\mathbf{z}}_r \\ \vec{\mathbf{z}}_{r'} \\ \vec{\mathbf{z}}_{sx} \\ \vec{\mathbf{z}}_{sx'} \\ \vec{\mathbf{z}}_2 \end{bmatrix} = \mathbf{A}' \vec{\mathbf{z}}' = \vec{\mathbf{t}}' \in \mathcal{R}_q^{t'}, \\
&\mathbf{u}'^\top \vec{\mathbf{z}}' + \langle \vec{\mathbf{z}}_r, \vec{\mathbf{z}}_{r'} \rangle + \langle \vec{\mathbf{z}}_{sx}, \vec{\mathbf{z}}_{sx'} \rangle = \mathbf{t}'', \\
&\mathrm{ct}(\sigma(\vec{\mathbf{e}}_i)^\top \vec{\mathbf{z}}_{r'} - \sigma(\vec{\mathbf{m}}_i)^\top \vec{\mathbf{z}}_r) = 0, \quad \forall i \in \{1, \ldots, \ell d\}, \\
&\mathrm{ct}(\sigma(\vec{\mathbf{e}}_j)^\top \vec{\mathbf{z}}_{sx'} - \sigma(\vec{\mathbf{m}}'_j)^\top \vec{\mathbf{z}}_{sx}) = 0, \quad \forall j \in \{1, \ldots, (m + \ell m')bd\}.
\end{aligned}$$

The parameter n' is chosen to simultaneously divide ℓ, $(m + \ell m')b$, and m_2, ensuring that all witness and constraint domains can be cleanly decomposed into n'-dimensional blocks. Accordingly, the witness can be represented as r vectors of length n', denoted as $\vec{\mathbf{s}}'_i$ for $i \in [r]$, which collectively cover all components. In the verification equations, we use vectors $\vec{\mathbf{z}}_i$, $\forall i \in \mathcal{Z}$, which are constructed by concatenating the corresponding $\vec{\mathbf{s}}'_i$ vectors. All verification equations are expressed in LaBRADOR 's principal relation form and hence we get the statement tuple, $\mathrm{x}_{pr} = \Big(\big((\mathbf{a}_i^{(k)})_{i \in [r]}, (\vec{\boldsymbol{\varphi}}_i^{(k)})_{i \in [r]}, \mathbf{b}^{(k)}\big)_{k \in [t'+1]}, \; \big((\mathbf{a}_i^{\prime(\ell)})_{i \in [r]}, (\vec{\boldsymbol{\varphi}}_i^{\prime(\ell)})_{i \in [r]}, b_0^{\prime(\ell)}\big)_{\ell \in [L]}\Big)$,
where $L = (\ell + (m + \ell m')b)d$.

LaBRADOR protocol. Let $\pi'_{\mathsf{LNP}} = (a_1, a_2, a_3, a_4)$ denote the output of the LNP layer without the final message a_5, x_{pr} denote the statement for the LaBRADOR relation and x denote the statement of the original ZK-LaBRADOR relation. We now define the protocol for proving principal relations below.

First round (Commit). The prover commits to each witness vector using an Ajtai commitment $\vec{\mathbf{u}}_i = \tilde{\mathbf{A}}\vec{\mathbf{s}}_i' \in \mathcal{R}_q^{\kappa},\ i \in [r]$, where κ is chosen such that $M\text{-}SIS_{\kappa,2\beta'}$ is hard, ensuring binding. These are referred to as *inner commitments*. Since each $\vec{\mathbf{u}}_i$ may not be short, it is decomposed into $t_1 \geq 2$ base-b_1 parts and concatenated into $\vec{\mathbf{u}} \in \mathcal{R}_q^{rt_1\kappa}$. To enable efficient batch openings, the prover additionally commits to *garbage polynomials* $\mathbf{g}'_{i,j} = \langle \vec{\mathbf{s}}_i', \vec{\mathbf{s}}_j' \rangle$ for $i, j \in [r]$. These polynomials are decomposed into t_2 base-b_2 parts and concatenated to form a single vector $\vec{\mathbf{g}}' \in \mathcal{R}_q^{((r^2+r)/2)t_2}$. Instead of sending *inner commitments* individually, the prover computes a single *outer commitment* to $(\vec{\mathbf{u}}, \vec{\mathbf{g}}')$, computed as $\vec{\mathbf{o}}_1 = \tilde{\mathbf{B}}\vec{\mathbf{u}} + \tilde{\mathbf{C}}\vec{\mathbf{g}}' \in \mathcal{R}_q^{\kappa_1}$. The prover outputs the message and corresponding challenge as

$$\mathfrak{a}_1 = ((\vec{\mathbf{z}_x}, \mathrm{x}_{pr}, \vec{\mathbf{o}_1}) \text{ and } \Pi_i = \mathrm{H}(1, \mathrm{crs}, \mathrm{x}, \pi'_{\mathsf{LNP}}, \mathfrak{a}_1) \in \mathsf{C}^{2\lambda \times (n'd)}.$$

Second round (Projection). The prover computes the projection vector as $\vec{p} = \sum_{i=1}^{r} \Pi_i \tau(\vec{\mathbf{s}}_i') \in \mathbb{Z}_q^{2\lambda}$. Let $\vec{\boldsymbol{\pi}}_i^{(j)}$ denote the ring element corresponding to the j-th row of Π_i. To ensure the correctness of the projection, for each $j \in [2\lambda]$, we have

$$0 = \sum_{i=1}^{r} \langle \tau(\vec{\boldsymbol{\pi}}_i^{(j)}), \tau(\vec{\mathbf{s}}_i') \rangle - p_j \quad = \mathrm{ct}\left(\sum_{i=1}^{r} \langle \sigma_{-1}(\vec{\boldsymbol{\pi}}_i^{(j)}), \vec{\mathbf{s}}_i' \rangle - p_j \right).$$

The prover's next message and corresponding challenges are $\mathfrak{a}_2 = (\vec{p})$ and $(\vec{\psi}^{(k)}, \vec{\mu}^{(k)}, \vec{\omega}^{(k)})_{k \in [K = \lceil \lambda / \log_2(q) \rceil]} = \mathrm{H}\left(1, \mathrm{crs}, \mathrm{x}, \pi'_{\mathsf{LNP}}, \mathfrak{a}_1, \mathfrak{a}_2\right) \in \mathbb{Z}_q^{ldK} \times \mathbb{Z}_q^{(m+\ell m')bdK} \times \mathbb{Z}_q^{2\lambda K}$.

Third round (Aggregating $\mathcal{F}'$ constraints). The prover aggregates all the dot product constraints in $\mathcal{F}'$ into a single constraint. For security parameter λ, the prover aggregates $\mathcal{F}'$ to $K = \lceil \lambda / \log_2(q) \rceil$ number of constant-term constraints using random linear combinations. For $k \in [K]$,

$$\begin{aligned} f''^{(k)}(\vec{\mathbf{s}}_1', \ldots, \vec{\mathbf{s}}_r') &= \sum_{i=1}^{ld} \psi_i^{(k)}\, \sigma(\vec{\mathbf{e}_i})^{\top} \vec{\mathbf{z}_{r'}} - \sigma(\vec{\mathbf{m}_i})^{\top} \vec{\mathbf{z}_r} + \sum_{i=1}^{(m+\ell m')bd} \mu_i^{(k)} \Big(\sigma(\vec{\mathbf{e}_j})^{\top} \vec{\mathbf{z}_{sx'}} \\ &\quad - \sigma(\vec{\mathbf{m}_j'})^{\top} \vec{\mathbf{z}_{sx}} \Big) + \sum_{j=1}^{2\lambda} \omega_j^{(k)} \Big(\sum_{i=1}^{r} \langle \sigma_{-1}(\vec{\boldsymbol{\pi}}_i^{(j)}), \vec{\mathbf{s}}_i' \rangle - p_j \Big), \\ &= \sum_{i,j=1}^{r} \boldsymbol{a}_{i,j}''^{(k)} \langle \vec{\mathbf{s}}_i', \vec{\mathbf{s}}_j' \rangle + \sum_{i=1}^{r} \langle \vec{\boldsymbol{\varphi}}_i''^{(k)}, \vec{\mathbf{s}}_i' \rangle - b_0''^{(k)}, \end{aligned}$$

where coefficients $\boldsymbol{a}_{i,j}''^{(k)}$, $\vec{\boldsymbol{\varphi}}_i''^{(k)}$, and $b_0''^{(k)}$ set accordingly. The prover also computes $\mathbf{b}''^{(k)} = \sum_{i,j=1}^{r} \boldsymbol{a}_{i,j}''^{(k)} \langle \vec{\mathbf{s}}_i', \vec{\mathbf{s}}_j' \rangle + \sum_{i=1}^{r} \langle \vec{\boldsymbol{\varphi}}_i''^{(k)}, \vec{\mathbf{s}_i'} \rangle$. The prover's next message and corresponding challenges are $\mathfrak{a}_3 = (\mathbf{b}''^{(k)})_{k \in [K]}$ and

$$(\vec{\boldsymbol{\alpha}}, \vec{\boldsymbol{\beta}}) = \mathrm{H}(1, \mathrm{crs}, \mathrm{x}, \pi'_{\mathsf{LNP}}, \mathfrak{a}_1, \mathfrak{a}_2, \mathfrak{a}_3) \in \mathcal{R}_q^{(t'+1)+(\lceil \lambda / \log_2(q) \rceil)}.$$

Fourth round (Aggregating $\mathcal{F}$ constraints). Finally, the prover aggregates the constraints in $\mathcal{F}$ and $f^{(k)}$ into a single dot product constraint F.

$$\begin{aligned} F(\vec{\mathbf{s}}_1', \ldots, \vec{\mathbf{s}}_r') &= \sum_{i=1}^{t'} \boldsymbol{\alpha}_i (\mathbf{a}_i'^{\top} \vec{\mathbf{z}}' - \vec{\mathbf{t}}_i') + \boldsymbol{\alpha}_{t'+1} \Big(\mathbf{u}'^{\top} \vec{\mathbf{z}}' + \langle \vec{\mathbf{z}_r}, \vec{\mathbf{z}_{r'}} \rangle + \langle \vec{\mathbf{z}_{sx}}, \vec{\mathbf{z}_{sx'}} \rangle - \mathbf{t}'' \Big) \\ &\quad + \sum_{k=1}^{K} \boldsymbol{\beta}_k f''^{(k)}(\vec{\mathbf{s}}_1', \ldots, \vec{\mathbf{s}}_r') = \sum_{i,j=1}^{r} \boldsymbol{a}_{i,j} \langle \vec{\mathbf{s}}_i', \vec{\mathbf{s}}_j' \rangle + \sum_{i=1}^{r} \langle \vec{\boldsymbol{\varphi}}_i, \vec{\mathbf{s}}_i' \rangle - \mathbf{b}, \end{aligned} \tag{7}$$

where $\boldsymbol{a}_{i,j}, \vec{\boldsymbol{\varphi}}_i$, and $\mathbf{b}$ are set accordingly.

The prover computes $\mathbf{h}'_{i,j} = \frac{1}{2}(\langle \vec{\boldsymbol{\varphi}}_i, \vec{\mathbf{s}}_j' \rangle + \langle \vec{\boldsymbol{\varphi}}_j, \vec{\mathbf{s}}_i' \rangle)$, which are then decomposed into t_1 base-b_1 parts and concatenated to form the garbage polynomial $\vec{\mathbf{h}}' \in \mathcal{R}_q^{((r^2+r)/2)t_1}$. Finally, the prover commits to $\vec{\mathbf{h}}'$ with $\vec{\mathbf{o}_2} = \tilde{\mathbf{D}}\vec{\mathbf{h}}' \in \mathcal{R}_q^{\kappa_2}$. The prover's next message and corresponding challenges are

$$\mathfrak{a}_4 = (\vec{\mathbf{o}_2}) \ \text{ and } (\mathbf{c}'_i)_{i \in [r]} = \mathrm{H}\left(1,\ \mathrm{crs},\ \mathrm{x}, \pi'_{\mathsf{LNP}}, \mathfrak{a}_1, \mathfrak{a}_2, \mathfrak{a}_3, \mathfrak{a}_4\right) \in \mathcal{C}^r.$$

Fifth round (Amortizing opening proof). In order to convince the verifier that $F(\vec{\mathbf{s}}_1', \ldots, \vec{\mathbf{s}}_r') = \mathbf{0}$, the prover provides a random linear combination of the witness vectors. The prover computes $\vec{\mathbf{z}} = \sum_{i=1}^{r} \mathbf{c}_i' \vec{\mathbf{s}}_i'$. The prover's last message in LaBRADOR protocol is $\mathfrak{a}_5 = \vec{\mathbf{z}}$.

Recursion layer. Instead of outputting the final message $\mathfrak{a}_5$ from the LaBRADOR protocol, the prover reuses it to define a new statement compatible with the principal relation with $\vec{\mathbf{u}}, \vec{\mathbf{g}}', \vec{\mathbf{h}}'$ and $\vec{\mathbf{z}}$ as witness vectors. Let $\vec{\mathbf{u}}' = \vec{\mathbf{u}} || \vec{\mathbf{g}}' || \vec{\mathbf{h}}' \in \mathcal{R}_q^{m'}$, with $m' = rt_1\kappa + (\frac{r^2+r}{2})(t_1 + t_2)$. To avoid a norm blowup, the prover decomposes the witness vector $\vec{\mathbf{z}}$ into $\vec{\mathbf{z}}^{(0)}$ and $\vec{\mathbf{z}}^{(1)}$. The new witness vectors are $\vec{\mathbf{z}}^{(0)}, \vec{\mathbf{z}}^{(1)}$, and $\vec{\mathbf{u}}'$. The original verification equations, which depend linearly on $\vec{\mathbf{u}}, \vec{\mathbf{g}}'$, and $\vec{\mathbf{h}}'$, are reformulated into dot product constraints F in terms of $\vec{\mathbf{z}}^{(0)}, \vec{\mathbf{z}}^{(1)}$ and $\vec{\mathbf{u}}'$. The norm checks for the commitment openings are combined into one global ℓ_2-norm bound, $\left\|\vec{\mathbf{z}}^{(0)}\right\|^2 + \left\|\vec{\mathbf{z}}^{(1)}\right\|^2 + \left\|\vec{\mathbf{u}}'\right\|^2 \leq \beta''^2$. Next, the witness is folded to obtain better parameters: $\vec{\mathbf{z}}^{(0)}, \vec{\mathbf{z}}^{(1)}$, and $\vec{\mathbf{u}}'$ are folded and combined into r' new witness vectors $\vec{\mathbf{w}}_1', \ldots, \vec{\mathbf{w}}_{r'}'$, which form the witness for the recursive statement $(\mathcal{F}, \beta'')$ in $\mathcal{R}$. The purpose of the recursion is to progressively reduce the witness rank n; this requires only $O(\log \log n)$ iterations of the base protocol.

Verification. We define the verification equations for the LNP layer followed by a single round of LaBRADOR. When the LaBRADOR layer is applied recursively, these equations are adapted in the same way at each recursion level. In the final round, the prover outputs $\vec{\mathbf{z}}, \vec{\mathbf{u}}, \vec{\mathbf{g}}', \vec{\mathbf{h}}'$. The verifier then checks:

1. $\|\vec{z_3}\| \overset{?}{\leq} \beta_3$.
2. If the following holds:
   (a) $ct(\mathbf{h}_i) \overset{?}{=} \sum_{j=1}^{256} \nu_{i,j} z_{3,j}, \quad \forall i \in [\lambda]$,
   (b) $(\vec{\mathbf{z}}, \vec{\mathbf{z}}) \overset{?}{=} \sum_{i,j=1}^{r} \mathbf{g}'_{i,j} \mathbf{c}_i \mathbf{c}_j$,
   (c) $\sum_{i=1}^{r} (\vec{\boldsymbol{\varphi}_i}, \vec{\mathbf{z}}) \mathbf{c}_i \overset{?}{=} \sum_{i,j=1}^{r} \mathbf{h}'_{i,j} \mathbf{c}_i \mathbf{c}_j$,
   (d) $\sum_{i,j=1}^{r} \boldsymbol{a}_{i,j} \mathbf{g}'_{i,j} + \sum_{i=1}^{r} \mathbf{h}'_{i,i} - \mathbf{b} \overset{?}{=} \mathbf{0}$ where
   $\boldsymbol{a}_{i,j} \overset{?}{=} \sum_{k=1}^{t'+1} \boldsymbol{\alpha}_k \mathbf{a}_{ij}^{(k)} + \sum_{k=1}^{\lceil \lambda / \log_2(q) \rceil} \boldsymbol{\beta}_k (\sum_{l=1}^{L} \psi_l^{(k)} \mathbf{a}_{ij}^{\prime(l)}) \quad \forall i, j \in [r]$ and $\mathbf{b} \overset{?}{=} \sum_{k=1}^{K} \boldsymbol{\alpha}_k \mathbf{b}^{(k)} + \sum_{k=1}^{\lambda / \log_2(q)} \boldsymbol{\beta}_k \mathbf{b}''^{(k)}$.
3. If $\vec{\mathbf{z}}$ is a valid (short) amortized opening to $\vec{\mathbf{z}}_i$.
4. If $(\vec{\mathbf{u}}, \vec{\mathbf{g}}')$ and $\vec{\mathbf{h}}'$ form valid (short) openings of $\vec{\mathbf{o}}_1$ and $\vec{\mathbf{o}}_2$, respectively.

Knowledge soundness of LaBRADOR. The knowledge soundness analysis of the LaBRADOR layer follows directly from the PSS framework of [1]. As shown therein, the LaBRADOR layer admits $(\mathbf{K}, \mathbf{R}, \boldsymbol{\Phi})$-coordinate-wise predicate-special-soundness, where $\mathbf{K} = (2, 2, 2, 3, \ldots, 2, 2, 2, 3)$, $\mathbf{R} = (1, 1, 1, r_1, \ldots, 1, 1, 1, r_t)$, $\boldsymbol{\Phi} = (\Phi_1^{\text{com}}, \Phi_2^{\text{com}}, \Phi_3^{\text{com}}, (\Phi_{4,1}^{\text{chal}}, \Phi_{4,2}^{\text{chal}}, \Phi_4^{\text{com}}), \ldots, \Phi_1^{\text{com}}, \Phi_2^{\text{com}}, \Phi_3^{\text{com}}, (\Phi_{4,1}^{\text{chal}}, \Phi_{4,2}^{\text{chal}}, \Phi_4^{\text{com}}))$, where for $i \in [t]$, r_i represents the number of witnesses at the i-th recursion level and t is the total number of recursions. Hence, our LaBRADOR layer inherits the predicate special soundness and knowledge extraction properties from the original construction, as detailed in Appendix B.

4.2 Security of Non-interactive ZK-LaBRADOR

In this section, we formally prove the knowledge soundness of $\Pi_{\text{M-SIS}}^{\text{NIZK}}$. We provide a brief intuition for the other properties: *completeness, zero-knowledge*, and *succinctness.*

Completeness. The honest executions of LNP and LaBRADOR satisfy their respective verification equations with overwhelming probability. In LNP, the completeness error comes from the rejection sampling step, and [7, Lemma 5.5] bounds this rejection rate, ensuring that the prover's messages are accepted except with negligible probability. In LaBRADOR, completeness error originates from the projection sampling step, and its success probability is bounded by Lemma 1. Since both subprotocols incur only negligible completeness error, the ZK-LaBRADOR protocol achieves negligible completeness error. The completeness will follow directly for the non-interactive ZK-LaBRADOR by the Fiat-Shamir transform.

Honest-Verifier Zero-knowledge (HVZK). This property follows directly from the LNP layer, which guarantees (non-abort) HVZK under the Extended-MLWE assumption [12, Def. 2.3], ensuring that the simulator generates transcripts that are indistinguishable from real ones for the non-abort executions.

Since LaBRADOR only relies on the simulator's public outputs, the combined ZK-LaBRADOR protocol preserves zero-knowledge. In the non-interactive setting, ZK-LABRADOR always generates non-abort transcripts; the same simulation strategy applies directly and hence HVZK for ZK-LaBRADOR follows immediately from its interactive counterpart.

Succinctness. In LNP, all messages except the prover's final response are succinct. By integrating this final message into the LaBRADOR layer, the resulting ZK-LaBRADOR construction retains overall succinctness.

4.2.1 Knowledge Soundness. In this section, we establish the knowledge soundness of the non-interactive ZK-LaBRADOR protocol. Based on the subextractor structure from Sec 3.2, which remains identical for the LaBRADOR layer and is extended accordingly, our analysis begins by demonstrating how a valid witness can be extracted from the accepting transcripts. Then we formalize the predicate-special-soundness of ZK-LaBRADOR for the relation formed using the extracted witnesses. We then define the *binding relation* underlying the ZK-LaBRADOR protocol, which enables us to formally establish knowledge soundness for the combined relation under the Fiat–Shamir transformation.

Extraction of witnesses. Consider the transcript tree of the composed protocol, where the interaction unfolds across multiple levels governed by the LaBRADOR and LNP components. For the extraction procedure, we identify two branching points where challenges are drawn from the special challenge space, one corresponding to the final level of the LaBRADOR component and the other to the final level of the LNP component.

For each transcript, the LaBRADOR extractor operates over the permissible set of challenge vectors. Let $\vec{\mathbf{c}_1} = \{\mathbf{c}_{1,i}\}_{i\in[r]}$ denote the challenge vector of transcript t_1. For each $i \in [r]$, the challenge vector of transcript t_2 is defined as $\vec{\mathbf{c}}_{2,i} = \{\mathbf{c}_{1,1}, \ldots, \mathbf{c}^{(2)}_{1,i}, \ldots, \mathbf{c}_{1,r}\}$. For all $i \in [r]$, the challenge vectors differ at exactly one point (i^{th} coordinate), enabling extraction of multiple witnesses. Using these pairs of challenges, we can compute the weak openings as

$$\bar{\mathbf{c}}_i = \mathbf{c}_{1,i} - \mathbf{c}_{2,i} \quad , \quad \vec{\mathbf{s}}'^{*}_i = \bar{\boldsymbol{c}}_i^{-1}\left(\vec{\boldsymbol{z}}\left(t_1\right) - \vec{\boldsymbol{z}}\left(t_{2,i}\right)\right) \forall i \in [r].$$

Due to the projection sampling step in the LaBRADOR layer, a relaxation factor $\sigma = \sqrt{\lambda/C^{(2)}}$ is introduced, i.e, $\sum_{i=1}^{r} \vec{\mathbf{s}}'^{*}_i \leq \sqrt{\lambda/C^{(2)}}\,\beta'$. Therefore, q must satisfy $q > \beta' C^{(1)}$. These bounds follow from the JL lemma 1. Finally, we can construct the witnesses of $\vec{\mathbf{z}_r}^{*}$, $\vec{\mathbf{z}_{r'}}^{*}$, $\vec{\mathbf{z}_{sx}}^{*}$, $\vec{\mathbf{z}_{sx'}}^{*}$, $\vec{\mathbf{z}_2}^{*}$ of the LaBRADOR by concatentating the corresponding $\vec{\mathbf{s}}'^{*}_i$ vectors.

Next, we construct the openings $(\vec{\mathbf{z}_1}^{*}, \vec{\mathbf{z}_2}^{*})$ for the LNP layer as

$$\vec{\mathbf{z}_1}^{*} = \begin{bmatrix} \vec{\mathbf{z}_x}^{*} \\ \vec{\mathbf{z}_r}^{*} \\ \vec{\mathbf{z}_{sx}}^{*} \end{bmatrix}, \vec{\mathbf{z}_2}^{*} = \vec{\mathbf{z}_2}^{*}$$

where $\vec{\mathbf{z}_x}^{*} = \vec{\mathbf{z}_x}$ is publicly known as a part of transcript. Then the same extraction procedure is applied to another set of transcripts t'_1 and t'_2.

For the LNP layer, we proceed the extraction by considering two accepting transcripts $\mathtt{t}_1$ and $\mathtt{t}_2$, where $\mathtt{t}_1$ and $\mathtt{t}_2$ consists of (t_1, t_2) and (t'_1, t'_2) transcripts respectively. Let $(\vec{\mathbf{z}_1}^{*}, \vec{\mathbf{z}_2}^{*})$ and $(\vec{\mathbf{z}_1}^{**}, \vec{\mathbf{z}_2}^{**})$ denote the LaBRADOR witnesses extracted from (t_1, t_2) and (t'_1, t'_2) respectively. Then, using two distinct challenges $\mathbf{c}_1$ and $\mathbf{c}_2$, we extract the original witnesses from the LNP layer as

$$\bar{\mathbf{c}} = \mathbf{c}_2 - \mathbf{c}_1, \qquad \vec{\mathbf{s}}^{*}_i = (\bar{\mathbf{c}})^{-1}(\vec{\mathbf{z}}^{**}_i - \vec{\mathbf{z}}^{*}_i)\,, \qquad \text{for } i \in \{1, 2\}.$$

Then, from the approximate range proof (ARP), the extracted vector $\vec{\mathbf{s}_1}^{*}$ satisfies $\|\vec{\mathbf{s}_1}^{*}\| \leq \beta_3 \cdot \frac{2}{\sqrt{26}}$. To ensure $\langle \vec{\mathbf{s}_1}^{*}, \vec{\mathbf{s}_1}^{*} - \vec{1} \rangle = 0$ over integers (i.e. no modular wrap-around), we require $\|\vec{\mathbf{s}_1}^{*}\|_1 \leq \|\vec{\mathbf{s}_1}^{*}\|_2 \sqrt{m_1 d}$. Setting $q > \beta_3^2 \frac{2}{\sqrt{26}} + 2\beta_3 \sqrt{m_1 d/26}$ ensures that

$$|\langle \vec{\mathbf{s}_1}^{*}, \vec{\mathbf{s}_1}^{*} - \vec{1} \rangle| \leq \|\vec{\mathbf{s}_1}^{*}\|_2^2 + \|\vec{\mathbf{s}_1}^{*}\|_2 \sqrt{m_1 d} < q.$$

Finally the vectors $\vec{\mathbf{x}}^{*}$, $\vec{\mathbf{r}}^{*}$, and $\vec{\mathbf{s}}'^{*}_x$ are recovered by splitting the concatenated vector $\vec{\mathbf{s}_1}^{*}$ by dimension.

Predicate Special Soundness of ZK-LABRADOR. Consider the ZK-LABRADOR protocol for relaton R

$$\mathrm{R} = \left\{ \left((\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}, \mathbf{B}, \vec{\mathbf{t}}, \vec{\mathbf{c}}), (\vec{\mathbf{x}}^*, \vec{\mathbf{r}}^*, \vec{\mathbf{s}}_x'^*)\right) \;\middle|\; \begin{array}{l} (\vec{\mathbf{x}}^*, \vec{\mathbf{r}}^*, \vec{\mathbf{s}}_x'^*) \in \mathcal{B}_1^{\ell+l+(m+\ell m')b} \\ \mathbf{A}_1 \vec{\mathbf{x}}^* + \mathbf{A}_2 \vec{\mathbf{r}}^* = \vec{\mathbf{t}} \\ \left[\mathbf{A} \mid \mathbf{B} - \vec{\mathbf{x}}^{*\top} \otimes \mathbf{G}\right] \mathbf{G}_\beta \cdot \vec{\mathbf{s}}_x'^* = \vec{\mathbf{c}} \\ \mathbf{x}_i^* \cdot (\mathbf{x}_i^* - 1) = 0 \quad \text{for } i \in [\ell]. \end{array} \right\}.$$

Now, we define the predicate-special-soundness property of the ZK-LaBRADOR protocol.

Lemma 9. *Adaption of [1, Lem.I.5]. Let Π denote the* ZK-LaBRADOR *protocol described in Sec 4. Define $\boldsymbol{K} = (2,2,2,3,2,2,2,3)$, $\boldsymbol{R} = (1,1,1,1,1,1,1,r_t)$, and let $\boldsymbol{\Phi}$ be the predicate system comprising the commitment predicates. Then, Π is $(\boldsymbol{K}, \boldsymbol{R}, \boldsymbol{\Phi})$-coordinate-wise predicate-special-sound for the relation* R*.*

Proof. This follows from the coordinate-wise predicate special soundness of the LNP (Lemma 8) and LaBRADOR components (Lemma 11). For the LNP layer, since all coordinates are fixed to 1, the CWPSS notion reduces to the standard predicate special soundness (PSS).

In the sequential composition of LNP followed by LaBRADOR, the combined predicate system $\boldsymbol{\Phi}$ integrates predicates from both the layers, with each level ensuring that the verification checks from the preceding level are preserved by enforcing them as predicates in the subsequent level.

Recursion: Let us consider the recursive composition of the LaBRADOR protocol, applied iteratively t times to compress the final proof size. Across these recursive iterations, each level similarly preserves the verification checks from the previous level. Let the sequence of t compositions be represented as a $(\boldsymbol{K}, \boldsymbol{R}, \boldsymbol{\Phi})$-coordinate-wise predicate-special-sound system, where

$$\boldsymbol{K} = (2,2,2,3,2,2,2,3,\ldots,2,2,2,3) \text{ and } \boldsymbol{R} = (1,1,1,1,1,1,1,r_1,\ldots,1,1,1,r_t).$$

The predicate system $\boldsymbol{\Phi}$ is then defined by the predicates from the LNP layer, together with the predicates recursively applied from the first iteration in the LaBRADOR layer,

$$\Phi_1^{\mathrm{com}}, \Phi_2^{\mathrm{com}}, \Phi_3^{\mathrm{com}}, (\Phi_{4,1}^{\mathrm{chal}}, \Phi_{4,2}^{\mathrm{chal}}, \Phi_4^{\mathrm{com}}), \ldots, \Phi_1^{\mathrm{com}}, \Phi_2^{\mathrm{com}}, \Phi_3^{\mathrm{com}}, (\Phi_{4,1}^{\mathrm{chal}}, \Phi_{4,2}^{\mathrm{chal}}, \Phi_4^{\mathrm{com}}).$$

Binding Relation. The LNP protocol has ABDLOP commitments which are paramterized by $\bar{\mathbf{A}} = [\bar{\mathbf{A}}_1 \mid \bar{\mathbf{A}}_2] \in \mathcal{R}_q^{n \times (m_1+m_2)}$. The LaBRADOR protocol involves one inner and two outer commitments — the inner commitments correspond to $\tilde{\mathbf{A}} \in \mathcal{R}_q^{\kappa \times n'}$, the first outer commitments are aggregated via $\tilde{\mathbf{B}}_i \in \mathcal{R}_q^{\kappa_1 \times \kappa}$ and $\tilde{\mathbf{C}}_{ijk} \in \mathcal{R}_q^{\kappa_2 \times 1}$. When $\kappa = \kappa_1 = \kappa_2$, these are merged into a single Ajtai commitment parameterized by $\tilde{\mathbf{E}} \in \mathcal{R}_q^{\kappa \times n_1}$ and the second outer commitment is parameterized by $\tilde{\mathbf{D}} \in \mathcal{R}_q^{\kappa \times n_2}$, where $n_1 = r\kappa + \frac{r^2+r}{2} k_2$, and $n_2 = \frac{r^2+r}{2} k_1$. Therefore the predicate system of the ZK-LaBRADOR admits the following binding relation:

$$\mathrm{R}_{\mathrm{pp}}^{\text{M-SIS}} = \mathrm{R}_{n, m_1+m_2, 4\eta(\beta_1^2+\beta_2^2), \bar{\mathbf{A}}}^{\text{M-SIS}} \cup \mathrm{R}_{\kappa, n, 8T_{\mathrm{op}}(b+1)\beta'', \tilde{\mathbf{A}}}^{\text{M-SIS}} \cup \mathrm{R}_{\kappa_1, n_1, 2\beta'', \tilde{\mathbf{E}}}^{\text{M-SIS}} \cup \mathrm{R}_{\kappa_1, n_2, 2\beta'', \tilde{\mathbf{D}}}^{\text{M-SIS}}.$$

Lemma 10. *Adaption of [1, Theorem.G.1]. Let Π be the base* ZK-LaBRADOR *protocol described in Sec.4 for the relation* R*. We consider its instantiation over a ring $\mathcal{R}_q$ of degree d, with a splitting factor of l, and a well-defined challenge space $\mathcal{C}$. The Fiat–Shamir transformation of* ZK-LaBRADOR *achieves knowledge soundness for the relation $\mathrm{R}_\sigma \cup \mathrm{R}_{pp}^{\text{M-SIS}}$ with the knowledge error*

$$2(Q+1)\left(\left(2^{-\lambda} + q^{-\lambda} + q^{-d/2} + 2lB\right) + \textstyle\sum_{i=1}^{t} \left(2^{-\lambda} + q^{-\lceil \lambda / \log q \rceil} + q^{-d/2} + (5+2l) r_i B\right)\right),$$

where Q denotes the number of adversarial queries and r_i is the number of witness vectors in iteration i. The expected number of extractor queries is at most $K + Q(K-1)$, where

$$K = 24 \cdot \prod_{i=1}^{t} (2^3 \cdot (2r_i + 1))$$

denotes the product of all challenges across the initial LNP *layer and the t recursive* LaBRADOR *iterations.*

Proof. The claim follows from the knowledge soundness of LNP established under the PSS framework (Theorem.2) and the knowledge soundness of LaBRADOR (Theorem.3). In particular, the norm checks in Φ_1^{com} within the LaBRADOR layer introduce a slack of $\sqrt{\lambda/C^{(2)}}$, which must be tolerated by the corresponding M-SIS bounds. The soundness error across the LNP layer and the t recursive iterations of the LaBRADOR layer accumulates additively, while the runtime grows multiplicatively with the number of recursion levels. Moreover, the sequence of witness ranks $r_1, \ldots, r_t$ decreases exponentially with the recursion depth, ensuring that only a logarithmic number of iterations are required.

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A LNP layer for Revocable Anonymous Credentials

This section describes the LNP layer of [9], which is used to prove the relation in Equation 4. In this layer, the witness vector $\vec{\mathbf{s}_x} \in \mathcal{R}_q^{m+\ell m'}$ is binary-decomposed into $\vec{\mathbf{s}_x'} \in \mathcal{B}_1^{(m+\ell m')b}$, where $b = \lceil \log \beta \rceil$. This is done to improve efficiency, as proving the exact norm relation $\langle \vec{\mathbf{s}_x}, \vec{\mathbf{s}_x} \rangle = \beta^2 \pmod q$ requires an impractically large modulus q due to the large value of β used in practice. By expressing $\vec{\mathbf{s}_x}$ in binary, the protocol efficiently proves that it is a binary vector while implicitly bounding its norm. Consequently, during extraction, the reconstructed witness $\vec{\mathbf{s}_x}$ has a norm bounded by $\beta\sqrt{(m+\ell m')d}$, which is sufficient to ensure the security of the accumulator under standard scaling arguments [11]. Hence the verification relation becomes

$$[\mathbf{A} \mid \mathbf{B} - \vec{\mathbf{x}}^\top \otimes \mathbf{G}]\mathbf{G}_\beta \cdot \vec{\mathbf{s}_x'} = \vec{\mathbf{c}}, \quad \text{where } \mathbf{G}_\beta = I_{m+\ell m'} \otimes \left[1\ 2 \cdots 2^{b-1}\right].$$

Input relation. Proving the relations in Eq.4 reduces to proving knowledge of $\vec{\mathbf{x}} \in \mathcal{R}_q^\ell$, $\vec{\mathbf{r}} \in \mathcal{R}_q^\ell$, and $\vec{\mathbf{s}_x'} \in \mathcal{R}_q^{(m+\ell m')b}$ such that

$$\begin{aligned}
&(\vec{\mathbf{x}}, \vec{\mathbf{r}}, \vec{\mathbf{s}_x'}) \in \mathcal{B}_1^{\ell+l+(m+\ell m')b}, \\
&\mathbf{A}_1\vec{\mathbf{x}} + \mathbf{A}_2\vec{\mathbf{r}} = \vec{\mathbf{t}}, \\
&\left[\mathbf{A} \mid \mathbf{B} - \vec{\mathbf{x}}^\top \otimes \mathbf{G}\right] \mathbf{G}_\beta \cdot \vec{\mathbf{s}_x'} = \vec{\mathbf{c}}, \\
&\mathbf{x}_i \cdot (\mathbf{x}_i - 1) = 0 \quad \text{for } i = 1, 2, \ldots, \ell.
\end{aligned}$$

LNP Protocol. Let H denote the random oracle used in the non-interactive protocol. We now describe the non-interactive LNP protocol to prove the input relation.

First round (Commit). Let $m_1 := \ell + l + (m + \ell m')\, b$, $\vec{\mathbf{s}_1} = \begin{bmatrix} \vec{\mathbf{x}} \\ \vec{\mathbf{r}} \\ \vec{\mathbf{s}_x}' \end{bmatrix} \in \mathcal{R}_q^{m_1}$ be the secret vector, and $\bar{\mathbf{A}}_1 \in \mathcal{R}_q^{n \times m_1}, \bar{\mathbf{A}}_2 \in \mathcal{R}_q^{n \times m_2}$, and $\bar{\mathbf{B}}_y \in \mathcal{R}_q^{\frac{256}{d} \times m_2}$ be the public matrices used for commitment. The prover samples $\vec{\mathbf{s}_2} \xleftarrow{\$} \mathcal{B}_1^{m_2}$ and commits to the secret vector $\vec{\mathbf{s}_1}$ as $\vec{\mathbf{t}_A} := \bar{\mathbf{A}}_1 \vec{\mathbf{s}_1} + \bar{\mathbf{A}}_2 \vec{\mathbf{s}_2}$. Next, the prover samples short masking vectors $\vec{\mathbf{y}_1} \xleftarrow{\$} D_{\mathfrak{s}_1}^{m_1}, \vec{\mathbf{y}_2} \xleftarrow{\$} D_{\mathfrak{s}_2}^{m_2}$ and $\vec{\mathbf{y}_3} \xleftarrow{\$} D_{\mathfrak{s}_3}^{256/d}$ and computes $\vec{\mathbf{w}} := \bar{\mathbf{A}}_1 \vec{\mathbf{y}_1} + \bar{\mathbf{A}}_2 \vec{\mathbf{y}_2} \in \mathcal{R}_q^n$, $\vec{\mathbf{t}_y} := \bar{\mathbf{B}}_y \vec{\mathbf{s}_2} + \vec{\mathbf{y}_3} \in \mathcal{R}_q^{256/d}$. Finally, the prover samples a garbage polynomial vector $\vec{\mathbf{g}} \leftarrow \{\mathbf{x} \in \mathcal{R}_q : ct(\mathbf{x}) = 0\}^{\lambda}$ and commits to it as $\vec{\mathbf{t}_g} := \bar{\mathbf{B}}_g \vec{\mathbf{s}_2} + \vec{\mathbf{p}}$.

The prover sets the first message, and the corresponding challenge as

$$a_1 := (\vec{\mathbf{t}_A}, \vec{\mathbf{t}_y}, \vec{\mathbf{t}_g}, \vec{\mathbf{w}}) \text{ and } (\Re_0, \Re_1) = \mathrm{H}(1, \mathrm{crs}, x, a_1) \in \{0,1\}^{256 \times m_1 d} \times \{0,1\}^{256 \times m_1 d}.$$

Second round (Approximate Range Proof). Given the challenge from the previous round, the prover computes $\vec{z_3} := \tau(\vec{\mathbf{y}_3}) + \Re\tau(\vec{\mathbf{s}_1}) \in \mathbb{Z}^{256}$, where $\Re = \Re_0 - \Re_1$ and applies rejection sampling to $\vec{z_3}$. If no abort occurs, the prover sets the second message, and the corresponding challenge as

$$a_2 := (\vec{z_3}) \text{ and } \quad (\nu_{i,j})_{i \in \lambda, j \in 257} = \mathrm{H}(1, \mathrm{crs}, x, a_1, a_2) \in \mathbb{Z}_q^{\lambda \times 257}.$$

Third round (Aggregating constaints over $\mathbb{Z}_q$). Firstly, we have to prove that $\vec{z_3}$ is well-formed. Let $\vec{\mathbf{r}_j} \in \mathcal{R}_q^{m_1}$ denote the j-th row of $\Re$ and $\vec{\mathbf{e}_j} \in \mathcal{R}_q^{256/d}$ denote the unit vector with a 1 in position j. By the automorphism property, for all $j \in [256]$ we have $\mathrm{ct}(\sigma(\vec{\mathbf{r}_j})^\top \vec{\mathbf{s}_1} + \vec{\mathbf{e}_j}^\top \vec{\mathbf{y}_3}) = z_{3,j}$. It remains to verify that $\vec{\mathbf{s}_1} = (\mathbf{s}_{1,1}, \ldots, \mathbf{s}_{1,m_1}) \in \mathcal{B}_1^{m_1}$ has small coefficients. As argued earlier, it suffices to prove $\langle \vec{\mathbf{s}_1}, \vec{\mathbf{s}_1} - \mathbf{1} \rangle = 0 \pmod q$, which for sufficiently large q holds over the integers. Equivalently, this means that

$$\mathrm{ct}\left(\sum_{i=1}^{m_1} \mathbf{s}_{1,i} \cdot \sigma\left(\mathbf{s}_{1,i} - \sum_{j=0}^{d-1} X^j \right) \right) = 0.$$

To prove these relations, for $i \in [\lambda]$, the prover computes the masked openings $(\mathbf{h}_1, \ldots, \mathbf{h}_\lambda) \in \mathcal{R}_q^\lambda$ to the garbage polynomials in $\vec{\mathbf{g}} = (\mathbf{g}_1, \cdots \mathbf{g}_\lambda)$ as

$$\mathbf{h}_i := \mathbf{g}_i + \sum_{j=1}^{256} \nu_{i,j} \left(\sigma(\vec{\mathbf{r}_j})^\top \vec{\mathbf{s}_1} + \vec{\mathbf{e}_j}^\top \vec{\mathbf{y}_3} \right) + \nu_{i,257} \sum_{k=1}^{m_1} \mathbf{s}_{1,k} \cdot \sigma\left(\mathbf{s}_{1,k} - \sum_{j=0}^{d-1} X^j \right).$$

Since each $\nu_{i,j}$ is an integer and $\mathrm{ct}(\mathbf{h}_i) = 0$, the constant coefficient of $\mathbf{h}_i$ simplifies to $\sum_{j=1}^{256} \nu_{i,j} z_{3,j}$, which the verifier can check directly.

The prover sets the third round message and the corresponding challenge as

$$a_3 := (\mathbf{h}_1, \ldots, \mathbf{h}_\lambda) \text{ and } (\vec{\gamma_1}, \vec{\gamma_2}, \vec{\gamma_3}, \vec{\gamma_4}) = \mathrm{H}\big(1, \mathrm{crs}, x, a_1, a_2, a_3\big) \in \mathcal{R}_q^t \times \mathcal{R}_q^{m' \cdot n} \times \mathcal{R}_q^\ell \times \mathcal{R}_q^\lambda.$$

Fourth round (Aggregating constraints over $\mathcal{R}_q$). The goal of the prover is to prove

$$\vec{\gamma_1}^\top \vec{\mathbf{t}} + \vec{\gamma_2}^\top \vec{\mathbf{c}} + \sum_{i=1}^{\lambda} \boldsymbol{\gamma}_{4,i} \mathbf{h}_i = U_1 + U_2 + U_3 + U_4, \tag{8}$$

where U_1, U_2, U_3, and U_4 are defined as

$$\begin{aligned}
U_1 &= \vec{\gamma_1}^\top (\mathbf{A}_1 \vec{\mathbf{x}} + \mathbf{A}_2 \vec{\mathbf{r}}), \\
U_2 &= \vec{\gamma_2}^\top [\mathbf{A} \mid \mathbf{B} - \vec{\mathbf{x}}^\top \otimes \mathbf{G}] \mathbf{G}_\beta \vec{\mathbf{s}_x}', \\
U_3 &= \sum_{i=1}^{\ell} \boldsymbol{\gamma}_{3,i} \mathbf{x}_i (\mathbf{x}_i - \mathbf{1}), \\
U_4 &= \sum_{i=1}^{\lambda} \boldsymbol{\gamma}_{4,i} \Big(\mathbf{g}_i + \sum_{j=1}^{256} \nu_{i,j} (\sigma(\vec{\mathbf{r}_j})^\top \vec{\mathbf{s}_1} + \vec{\mathbf{e}_j}^\top \vec{\mathbf{y}_3}) + \nu_{i,257} \vec{\mathbf{s}_1}^\top \sigma(\vec{\mathbf{s}_1} - \mathrm{PW}) \Big),
\end{aligned}$$

and PW denotes the polynomial vector $\left(\sum_{j=0}^{d-1} X^j, \ldots, \sum_{j=0}^{d-1} X^j\right) \in \mathcal{R}_q^{m_1}$.

The intuition for proving a single quadratic equation is as follows. We fix a challenge $\mathbf{c} \in \mathcal{C}$ and define

$$\vec{\mathbf{z}_1} = \vec{\mathbf{y}_1} + \mathbf{c}\vec{\mathbf{s}_1} \left(\text{which can be parsed as } \begin{bmatrix} \vec{\mathbf{z}_x} \\ \vec{\mathbf{z}_r} \\ \vec{\mathbf{z}_{sx}} \end{bmatrix} \right) \quad \text{and} \quad \vec{\mathbf{z}_2} := \vec{\mathbf{y}_2} + \mathbf{c}\vec{\mathbf{s}_2},$$

which are later sent to the verifier in response to the challenge. The verifier thus can compute U once the vectors $\vec{\mathbf{z}_1}$ and $\vec{\mathbf{z}_2}$ are revealed and it is thus sufficient to show that the quadratic component of U vanishes to prove Eq. 8. U is computed as $U = \sum_{i=1}^{4} U_i(\vec{\mathbf{z}_1}, \vec{\mathbf{z}_2})$, where

$$\begin{aligned}
U_1(\vec{\mathbf{z}_1}) &= \mathbf{c}\vec{\boldsymbol{\gamma}_1}^\top(\mathbf{A}_1\vec{\mathbf{z}_x} + \mathbf{A}_2\vec{\mathbf{z}_r}),\\
U_2(\vec{\mathbf{z}_1}) &= \vec{\boldsymbol{\gamma}_2}^\top[\mathbf{cA} \mid \mathbf{cB} - \vec{\mathbf{z}_x}^\top \otimes \mathbf{G}]\,\mathbf{G}_\beta\vec{\mathbf{z}_{sx}},\\
U_3(\vec{\mathbf{z}_1}) &= \sum_{i=1}^{\ell} \boldsymbol{\gamma}_{3,i}\,\mathbf{z}_{x,i}(\mathbf{z}_{x,i} - \mathbf{c}),\\
U_4(\vec{\mathbf{z}_1}, \vec{\mathbf{z}_2}) &= \sum_{i=1}^{\lambda} \boldsymbol{\gamma}_{4,i}(\mathbf{c}\vec{\mathbf{e}_i}'^\top(\mathbf{c}\vec{\mathbf{t}_g} - \mathbf{B}_p\vec{\mathbf{z}_2}) + \mathbf{c}\sum_{j=1}^{256} \nu_{i,j}(\sigma(\vec{\mathbf{r}_j})^\top\vec{\mathbf{z}_1} + \vec{\mathbf{e}_j}^\top(\mathbf{c}\vec{\mathbf{t}_y} - \mathbf{B}_y\vec{\mathbf{z}_2}))\\
&\quad + \nu_{i,257}\vec{\mathbf{z}_1}^\top\sigma(\vec{\mathbf{z}_1} - \mathbf{c}\mathrm{PW})),
\end{aligned}$$

where $\vec{\mathbf{e}_i}' \in \mathcal{R}_q^{\lambda}$ denotes the unit vector with 1 exactly on the i-th position.

On expanding $\vec{\mathbf{z}_1}$ and $\vec{\mathbf{z}_2}$, the term U simplifies to $\mathbf{c}^2\sum_{i=1}^{4} U_i + \mathbf{c}\mathbf{f}_1 + \mathbf{f}_0$. The goal is to verify that $\sum_{i=1}^{4} U_i = 0$. However, the expansion introduces auxiliary terms $\mathbf{f}_0$ and $\mathbf{f}_1$ defined as follows:

$$\begin{aligned}
\mathbf{f}_1 &= \vec{\boldsymbol{\gamma}_1}^\top(\mathbf{A}_1\vec{\mathbf{y}_x} + \mathbf{A}_2\vec{\mathbf{y}_r}) + \vec{\boldsymbol{\gamma}_2}^\top([\mathbf{A} \mid \mathbf{B} - \vec{\mathbf{x}}^\top \otimes \mathbf{G}]\mathbf{G}_\beta\vec{\mathbf{y}_{sx}} - [0 \mid \vec{\mathbf{y}_x}^\top \otimes \mathbf{G}]\mathbf{G}_\beta\vec{\mathbf{s}_x})\\
&\quad + \sum_i \boldsymbol{\gamma}_{3,i}\mathbf{y}_{x,i}(2\mathbf{x}_i - 1) + \sum_i \boldsymbol{\gamma}_{4,i}\Big(- \mathbf{B}_g\vec{\mathbf{y}_2} + \sum_j \nu_{i,j}(\sigma(\vec{\mathbf{r}_j})^\top\vec{\mathbf{y}_1} - \vec{\mathbf{e}_j}^\top\mathbf{B}_y\vec{\mathbf{y}_2})\\
&\quad + \nu_{i,257}(\vec{\mathbf{y}_1}^\top\sigma(\vec{\mathbf{s}_1} - \mathrm{PW}) + \vec{\mathbf{s}_1}^\top\sigma(\vec{\mathbf{y}_1}))\Big),\\
\mathbf{f}_0 &= [0 \mid -\vec{\mathbf{y}_x}^\top \otimes \mathbf{G}]\mathbf{G}_\beta\vec{\mathbf{y}_{sx}} + \sum_i \boldsymbol{\gamma}_{3,i}\mathbf{y}_{x,i}^2 + \vec{\mathbf{y}_1}^\top\sigma(\vec{\mathbf{y}_1}).
\end{aligned}$$

The prover commits to $\mathbf{f}_1$, since it depends on witness vectors, as $\mathbf{t}_{\text{quad}} := \overline{\mathbf{B}}_{\text{quad}}\vec{\mathbf{s}_2} + \mathbf{f}_1$, and defines $\mathbf{v} := \overline{\mathbf{B}}_{\text{quad}}\vec{\mathbf{y}_2} + \mathbf{f}_0$, where $\overline{\mathbf{B}}_{\text{quad}} \in \mathcal{R}_q^{1\times m_2}$. Finally, the verifier checks $U - \left(\mathbf{c}\mathbf{t}_{\text{quad}} - \overline{\mathbf{B}}_{\text{quad}}\vec{\mathbf{z}_2}\right) + \mathbf{v} = 0$, which confirms that $\sum_{i=1}^{4} U_i = 0$.

The prover sets the fourth message and the corresponding challenge as

$$a_4 := (\mathbf{t}_{\text{quad}}, \mathbf{v}) \quad \text{and} \quad \mathbf{c} = \mathrm{H}(1,\ \mathrm{crs},\ \mathrm{x}, a_1, a_2, a_3, a_4) \in \mathcal{C}.$$

Fifth round (Opening). The prover computes $\vec{\mathbf{z}_i} = \vec{\mathbf{y}_i} + \mathbf{c}\vec{\mathbf{s}_i}$ for $i = 1, 2$ and applies rejection sampling for both $\vec{\mathbf{z}_1}$ and $\vec{\mathbf{z}_2}$. If no abort occurs, the prover outputs $\mathbf{z}_1, \mathbf{z}_2$. The prover's last message is $a_5 = (\vec{\mathbf{z}_1}, \vec{\mathbf{z}_2})$.

Verification. Finally, the verifier checks the following:

$$\begin{aligned}
&\|\vec{\mathbf{z}_1}\| \le \beta_1, \quad \|\vec{\mathbf{z}_2}\| \le \beta_2, \quad \|\vec{z_3}\| \le \beta_3,\\
&\bar{\mathbf{A}}_1\vec{\mathbf{z}_1} + \bar{\mathbf{A}}_2\vec{\mathbf{z}_2} = \vec{\mathbf{w}} + \mathbf{c}\vec{\mathbf{t}_A},\\
&U = \mathbf{c}^2\left(\vec{\boldsymbol{\gamma}_1}^\top\vec{\mathbf{t}} + \vec{\boldsymbol{\gamma}_2}^\top\vec{\mathbf{c}} + \sum_{i=1}^{\lambda}\boldsymbol{\gamma}_{4,i}\mathbf{q}_i\right) + \left(\mathbf{c}\mathbf{t}_{\text{quad}} - \bar{\mathbf{B}}_{\text{quad}}\vec{\mathbf{z}_2}\right) + \mathbf{v},\\
&\mathrm{ct}(\mathbf{h}_i) = \sum_{j=1}^{256}\nu_{i,j}z_{3,j}, \quad i = 1, \ldots, \lambda,
\end{aligned}$$

where $\beta_1 := \mathfrak{s}_1\sqrt{2m_1d}, \beta_2 := \mathfrak{s}_2\sqrt{2m_2d}$ and $\beta_3 := 1.7\mathfrak{s}_3\sqrt{256}$ (See [7, Lem. 5.5]).

B Predicate Special Soundness of LaBRADOR

We consider the LaBRADOR layer in Sec 4.1, whose PSS properties follow directly from the analysis presented in [1]. The principal relation of LaBRADOR is modified to include a norm slack parameter σ, defined as

$$R_\sigma = \left\{ ((\mathcal{F}, \mathcal{F}', \beta), (\vec{\mathbf{w}}_1^*, \ldots, \vec{\mathbf{w}}_r^*)) \;\middle|\; \begin{array}{l} \forall f \in \mathcal{F},\ f(\vec{\mathbf{w}}_1^*, \ldots, \vec{\mathbf{w}}_r^*) = 0, \\ \forall f' \in \mathcal{F}',\ \mathrm{ct}(f'(\vec{\mathbf{w}}_1^*, \ldots, \vec{\mathbf{w}}_r^*)) = 0 \bmod q, \\ \sum_{i=1}^r \|\vec{\mathbf{w}}_i^*\|_2^2 \le \sigma\beta^2 \end{array} \right\}.$$

Lemma 11. *Let Π be the LaBRADOR protocol with parameters $\mathbf{K} = (2,2,2,3)$, $\mathbf{R} = (1,1,1,r)$, and predicate system $\mathbf{\Phi}$ comprising $\Phi_1^{\mathrm{com}}, \Phi_2^{\mathrm{com}}, \Phi_3^{\mathrm{com}}, \Phi_4^{\mathrm{com}}$ and challenge predicates $\Phi_{4,1}^{\mathrm{chal}}, \Phi_{4,2}^{\mathrm{chal}}$. Then Π is $(\mathbf{K}, \mathbf{R}, \mathbf{\Phi})$-coordinate-wise predicate-special-sound for the relation R_σ with $\sigma = \sqrt{\lambda/C^{(2)}}$.*

Binding Relation. The LaBRADOR protocol employs one inner and two outer commitments. The inner commitment is parameterized by $\tilde{\mathbf{A}} \in \mathcal{R}_q^{\kappa \times n'}$. The first outer commitment is parameterised by $\tilde{\mathbf{B}}_i \in \mathcal{R}_q^{\kappa_1 \times \kappa}$ and $\tilde{\mathbf{C}}_{ijk} \in \mathcal{R}_q^{\kappa_2 \times 1}$. For $\kappa = \kappa_1 = \kappa_2$, these are merged into a single Ajtai commitment defined by $\tilde{\mathbf{E}} \in \mathcal{R}_q^{\kappa \times n_1'}$. The second outer commitment is parameterized by $\tilde{\mathbf{D}} \in \mathcal{R}_q^{\kappa \times n_2'}$, where $n_1' = r\kappa + \frac{r^2+r}{2}k_2, \quad n_2' = \frac{r^2+r}{2}k_1$.

When binding fails, extraction corresponds to solving the M-SIS relation,

$$\mathrm{R}^{\text{M-SIS}}_{\kappa,n,\beta^*,\mathbf{A}} = \left\{ \vec{\mathbf{v}} \in \mathcal{R}_q^n \mid \mathbf{A}\vec{\mathbf{v}} = \vec{\mathbf{0}},\ 0 < \|\vec{\mathbf{v}}\| \le \beta^* \right\}.$$

Accordingly, the predicate system of LaBRADOR admits the binding relation as follows

$$\mathrm{R}^{\text{M-SIS}}_{\text{pp}} = \mathrm{R}^{\text{M-SIS}}_{\kappa,n,8T_{\text{op}}(b+1)\beta'',\tilde{\mathbf{A}}} \cup \mathrm{R}^{\text{M-SIS}}_{\kappa_1,n_1',2\beta'',\tilde{\mathbf{E}}} \cup \mathrm{R}^{\text{M-SIS}}_{\kappa_1,n_2',2\beta'',\tilde{\mathbf{D}}}.$$

Theorem 3 (Knowledge Soundness). *Let Π be the base LaBRADOR protocol instantiated over a ring $\mathcal{R}_q$ of degree d with splitting factor l and a B-well-spread challenge set, where each challenge has operator norm at most T_{op}. The statements in the LaBRADOR relation are restricted such that ℓ_2-norm is at most $q/C^{(1)}$, where $C^{(1)}, C^{(2)}$ are parameters from Lemma 1. Then, the Fiat-Shamir transformation of LaBRADOR is knowledge-sound for $R_\sigma \cup R_{\text{pp}}^{\text{M-SIS}}$ with $\sigma = \sqrt{\lambda/C_2}$ and knowledge error*

$$2(Q{+}1)\Big(2^{-\lambda} + q^{-\lceil \lambda/\log q \rceil} + q^{-d/l} + (5{+}2l)rB\Big),$$

where the extractor makes at most $K + Q(K{-}1)$ queries in expectation, for $K = 2^3(2r{+}1)$, and the prover makes at most Q oracle queries.

Recursive Composition. When LaBRADOR is composed recursively t times, the prover commits to $(\vec{\mathbf{z}}, \vec{\mathbf{u}}, \vec{\mathbf{g}}', \vec{\mathbf{h}}')$ of one iteration as the first message of the next, leaving only the final opening. An accepting transcript of one iteration ensures soundness of checks by an honest verifier and in recursive composition, these guarantees propagate to the subsequent predicates. A sequence of t such compositions may therefore be represented as a $(\mathbf{K}, \mathbf{R}, \mathbf{\Phi})$-coordinate-wise predicate-special-sound protocol, where

$$\mathbf{K} = (2,2,2,3,\ldots,2,2,2,3), \text{ and } \mathbf{R} = (1,1,1,r_1,\ldots,1,1,1,r_t).$$

The predicate system $\mathbf{\Phi}$ is applied iteratively,

$$\Phi_1^{\mathrm{com}},\ \Phi_2^{\mathrm{com}},\ \Phi_3^{\mathrm{com}},\ (\Phi_{4,1}^{\mathrm{chal}}, \Phi_{4,2}^{\mathrm{chal}}, \Phi_4^{\mathrm{com}}), \ldots, \Phi_1^{\mathrm{com}},\ \Phi_2^{\mathrm{com}},\ \Phi_3^{\mathrm{com}},\ (\Phi_{4,1}^{\mathrm{chal}}, \Phi_{4,2}^{\mathrm{chal}}, \Phi_4^{\mathrm{com}}).$$

Each Φ_1^{com} enforces a norm slack $\sqrt{\lambda/C_2}$, hence the M-SIS norms must accommodate this factor. The cumulative soundness error over t iterations is

$$2(Q{+}1)\sum_{i=1}^{t}\Big(2^{-\lambda} + q^{-\lceil \lambda/\log q \rceil} + q^{-d/l} + (5{+}2l)r_iB\Big),$$

and the runtime scales multiplicatively, giving $K + Q(K{-}1)$ queries for $K = \prod_{i=1}^t \big(2^3(2r_i{+}1)\big)$. The sequence $r_1, \ldots, r_t$ decreases exponentially.